Practical Post-Quantum Secure Publicly Verifiable Secret Sharing and Applications*

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Abstract

We present a new framework for constructing practically efficient publicly verifiable secret sharing (PVSS) with *non-interactive dealers*, in that the dealer may go offline after sending a single message, and is not involved in the share verification process. We use identity-based encryption (IBE) and commitments as the main ingredients and avoid expensive zero-knowledge proofs. Instantiating them with post-quantum secure schemes, a lattice-based IBE and a hash-based commitment, we obtain our first construction – a post-quantum secure PVSS with non-interactive dealers that outperform the prior lattice-based practical construction, Gentry et al. [Eurocrypt 2022] by two orders of magnitude.

However, to enable the aggregation of PVSS transcripts (which facilitates many additional applications such as secure voting), we propose our second construction by replacing hash-based commitments with Pedersen's homomorphic commitments. While this does not achieve full-fledged post-quantum security (as Pedersen's scheme is not quantum safe), it still provides privacy against a post-quantum adversary. We prove the security of this construction in a new model, which we call long-lasting security. This model guarantees that the protocol is secure in the present (pre-quantum era) while maintaining privacy in the long term (post-quantum era). This new model is of independent interest for constructing efficient schemes that are resilient to harvest-now-decrypt-later line of attacks. In this model, we propose a blockchain-compatible secure voting scheme using our PVSS scheme.

Our PVSS schemes demonstrate practical efficiency: our post-quantum PVSS shares a secret among 1024 receivers in 692 ms and verifies the dealing in 128 ms, and communicates 4MB, overall yielding a two orders of magnitude improvement over the state of the art [Gentry et al., Eurocrypt 2022].

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1 Introduction

Publicly verifiable secret sharing (PVSS) schemes [33] enable a dealer to share a secret s among n receivers (typically using Shamir's secret sharing [56]) in a publicly verifiable manner, such that anyone, even outside the system, can verify that the sharing has been done correctly. Any $t+1$ ($t < n$) receivers can reconstruct the secret, while it stays hidden from any (possibly colluding) subset of t parties. PVSS has found numerous applications in the security space, including distributed key generation [11], threshold cryptosystems [30], secure voting [55], private auctions [59], and secure randomness on blockchain [46]. In particular, the public verifiability feature has emerged as one of the most crucial requirements for blockchain-based systems as it allows any potential user to ascertain the system's integrity.

In PVSS applications such as secure voting or secure auctions on blockchain, the protocol transcript is stored forever on the blockchain. In such cases, post-quantum security is crucial due to the ease of executing *harvest-now-decrypt-later* attack [15, 16, 31][1] on the transcript. For example, votes cast now privately, if they get leaked in a decade with a plausible quantum attack, that could have significant repercussions for voters – such a possibility poses a pragmatic resistance for large-scale adaptability of any such applications today.

Moreover, for many client-server applications, including those that can be built over blockchains, such as ZK-as-a-service [35], MPC-as-a-service [14, 58, 60], private statistics [8] etc., it is important to keep the dealer/client non-interactive – the dealer posts/sends its message, and then can go offline. In particular, in order to produce a (publicly verifiable) correctness proof, the dealer should not be involved at all. We say that such PVSS has a non-interactive dealer.

[1] In this attack, the adversary collects/harvests the encrypted data available presently. Once quantum computers become practical, it can then break the encryption security to extract the plaintexts. Since all data on blockchains is essentially always available, one can easily harvest a massive quantity of data from blockchains to prepare for such attacks.

With growing applications in the blockchain space, a few PVSS schemes has been proposed recently [24, 27, 40, 43, 45]. However, [4, 49] [2] are two proposals that are both *post-quantum secure* and have a *non-interactive dealer* follow a standard paradigm [3]: (i) given individual public keys for every receiver, the dealer encrypts their corresponding share with them plus also commits to the actual secret s; (ii) along with that it sends a non-interactive zero knowledge (NIZK) proof of correctness attesting to the fact that the encryptions indeed contains a t out of n Shamir shairng of the committed s. Clearly, as long as all the ingredients, namely, encryption, commitments, and NIZK are post-quantum secure, the scheme satisfies post-quantum security. Furthermore, the presence of NIZK ensures that dealer's message is already publicly verifiable without any additional step, thereby achieving the dealer's non-interactivity feature. Unfortunately, the final scheme incurs substantial overhead due to the use of post-quantum secure (lattice-based) NIZK proofs. This motivates us to ask the following question: p

Can we construct a practically-efficient post-quantum secure PVSS with a non-interactive dealer without expensive NIZK?

We answer the question in the affirmative by proposing a new approach to obtaining PVSS with non-interactive dealers.

1.1 Our Contributions

A new approach to PVSS with non-interactive dealer: We use a post-quantum secure identity-based encryption scheme (IBE) as the core tool. Each receiver-i publishes an individual master IBE public key mpk_i for which they store a master secret key msk_i. The dealer posts a commitment to each receiver's share $[\![s]\!]_i$ as $\mathsf{Com}(s_i)$, and encrypts each receiver's share $[\![s]\!]_i$ (and opening randomness) under index i and the dealer's identity id_d using mpk_i and then goes offline. The receivers check that the posted commitments commit to a degree t sharing of a polynomial. Then the i-th receiver, once decrypts the share and opening randomness, checks with the commitment $\mathsf{Com}([\![s]\!]_i)$ whether the share it obtained is correct – if not, then the receiver uses msk_i to extracts dealer's decryption key $\mathsf{sk}_{i,\mathsf{id}_d}$ – using this, anyone (even someone who is neither the dealer nor a receiver) can decrypt the ciphertext and can verify whether the receiver's claim is correct by matching with the commitment. Firstly, this clearly does not involve a NIZK proof; and secondly, the IBE enables the dealer to remain non-interactive, as the receiver can extract the dealer's decryption key locally using her master secret key msk_i.

Practically efficient PQ secure PVSS with non-interactive dealer. In our first construction, we use a lattice-based IBE [6, 32] and hash-based commitments to obtain overall post-quantum security – this yields a post-quantum secure PVSS with a non-interactive dealer which is around *two orders of magnitude* faster over state–of-the-art lattice based PVSS (viz. Gentry al. [37]).[4] Considering 1024 receivers, our PVSS protocol takes 692 ms to deal the secret and 128 ms to verify the shares, whereas [37] takes roughly 34 minutes to perform the same task. Our PVSS communicates 4MB to perform the dealing, whereas [37] communicates 251MB, yielding two orders of maagnitude improvement in communication size.

A new notion: Long-lasting security However, the transcripts in our PVSS protocol is not aggregatable (as defined in [12]) due to the use of hash-based commitments, which are not homomorphic. As a result, we cannot use our hash-based PVSS protocol for applications like secure voting, DKG or secure data aggregation as they require aggregatable protocol transcripts. To obtain such aggregatable PVSS protocols, we relax the security notion. Specifically, we introduce the notion of *long-lasting security*, which provides full classical computational security during the execution, and provides post-quantum privacy – this notion, while weaker than full-fledged post-quantum security, suffices to protect against harvest-now-decrypt-later attacks [15]. For example, in our secure voting application, while the integrity/validity of votes is guaranteed at present, when quantum computers are not available, the votes would remain private even in a post-quantum world.

Practically efficient long-lasting secure PVSS with non-interactive dealer. In this long-lasting setting, we propose our second PVSS construction by replacing the hash-based commitments with homomorphic Pedersen commitments (and adequately changing the verification procedure). Sharing a secret among 1024 receivers requires 867 ms, verifying a share requires 2.13 s, and the entire dealing requires 4MB communication.[5]

Secure Voting on Blockchain with long-lasting security. Using our aggregatable PVSS, we build a practical, secure voting protocol over blockchains in a client-server model. The entire protocol is publicly verifiable by any external party and provides long-lasting vote privacy, i.e. the vote of the client remains *private* even against a future quantum adversary. Conducting a poll with 128 servers and four poll options: client-side voting takes 0.6 s, vote verification takes 18 ms, and 100K votes are aggregated and decrypted in 62 ms - offering a 40× speedup over prior lattice-based voting schemes [19, 29].

[2] [37] achieves post-quantum privacy and not full post-quantum security due to reliance on Discrete-Log based Bulletproofs [20].

[3] The same paradigm is followed by many works including [33, 50, 55], albeit in the classical/pre-quantum setting.

[4] [37] does not achieve post-quantum security as mentioned in Table 1.

[5] Our notion of long-lasting security is inspired by the existing notion of everlasting security [61], in that the privacy is unconditional as opposed to post-quantum. Nevertheless, to instantiate any everlasting secure protocol [9] one needs to rely on secure channels, which must rely on computational assumptions, effectively rendering the overall protocol to our setting, if post-quantum secure channels are used. Our protocols do not rely on secure channels, but use authenticated bulletin boards or the blockchain.

We prove the security of all our protocols assuming an honest majority and synchronous communication channels. Our security proofs are in the classical random oracle model.

1.2 Related Works

We compare our PQ-secure and long-lasting secure PVSS protocols with existing PVSS protocols from the literature. A qualitative comparison can be found in Table. 1.

Discrete Log-based protocols: In traditional discrete log-based protocols [26, 33, 36, 50, 55], the dealer shares a secret using Shamir's scheme, publishing commitments and encrypted shares. Some protocols [36, 50] allow receivers to complain about inconsistent shares, requiring dealer interaction for resolution. These schemes are not publicly verifiable unless receivers add an extra multisignature round. The works of [26, 33, 55] make the dealer non-interactive by publishing NIZK proofs that each encrypted share matches the secret polynomial. While verifiable, these protocols are not post-quantum secure due to reliance on DL, and replacing DL-based components with lattice-based ones is impractical due to the high cost of lattice-based NIZK proofs, preventing efficient post-quantum or long-lasting security.

Class-group-based protocols: The works of [24, 43] construct non-interactive PVSS using class group encryptions, with NIZK proofs ensuring correct encryption. Receivers verify the proofs and decrypt their shares. However, class group encryptions are not known to be post-quantum secure, and the NIZKs are customized for class group encryptions; hence, long-lasting secure protocols are hard to construct.

Lattice-based VSS protocols: The recent work of [4] constructed a post-quantum non-interactive PVSS from vector commitments that satisfy linear opening, functional hiding, and NIZK proof of smallness properties, and an encryption scheme that supports efficient proofs of correct decryption. Concurrently, the work of [49] constructed a non-interactive PVSS from an encryption and a NIZK proof system for gap language. However, the practicality of these works is unclear as they lack a concrete implementation and both of them rely on expensive NIZK proofs over lattices. The work of Gentry et al. [37] designed a PVSS scheme that uses an LWE-based encryption scheme and uses DL-based Bulletproofs [20] to prove that the ciphertext is well-formed. Using Bulletproofs improved efficiency over the pure lattice-based solutions, but it failed to provide practical efficiency, as it takes around 34 minutes for the dealer to deal a single secret among 1024 receivers, whereas our work performs the dealing and verification in less than a second, yielding over three orders of magnitude improvement. Moreover, due to reliance on Bulletproofs, the PVSS of [37] achieves long-lasting security and not post-quantum security.

Hash-based protocols: The work of [10] introduced an efficient hash-based VSS where the dealer non-interactively distributes shares using hash-based distributed ZK proofs. However, during share verification, receivers can file complaints, requiring the dealer to reveal corresponding shares—making the protocol interactive and its transcript not publicly verifiable. A later improvement by [13] proposed a general framework for computational VSS in the honest-majority setting using Shamir secret sharing, random oracles, and non-homomorphic commitments. Yet, both [10] and the post-quantum secure variants of [13] require dealer interaction in the verification phase. In contrast, our post-quantum secure PVSS removes this interaction by leveraging IBE and thereby enables many new applications.

Bivariate-Polynomial Based: The works of [5, 7, 47] construct asynchronous VSS using bivariate polynomials, where the dealer commits to a random low-degree polynomial and distributes partial evaluations with proofs to nodes, who then collaborate to reconstruct shares. These protocols avoid PKI but require $O(n^2)$ exponentiations and communication, and rely on q-SDH commitments, making them insecure against quantum adversaries. Long-lasting security and public verifiability are possible from them using perfectly hiding commitments, lattice-based encryption for channels, and multisignature rounds, though at the cost of higher dealer computation.

Signature-based VSS protocols: The recent work by [27] constructs an asynchronous dual threshold VSS protocol using Polynomial commitments [44] and signatures where the dealer shares the secret among n receivers and waits for signed confirmations from the receivers. Once the dealer receives confirmations from $t+1$ receivers it broadcasts the shares of the rest of the $n-t-1$ receivers and the $t+1$ signatures. This enables efficient reconstruction later on. However, their dealer is interactive in nature. We observe that this VSS protocol can be made long-lasting secure (albeit with an interactive dealer) if the underlying Polynomial commitment is instantiated using the Pedersen variant of KZG commitment, as it provides computational binding and perfect hiding, and the point-to-point channels are replaced by a PQ-secure public key encryption.

Related works of Long-lasting Security and Voting: The notion of long-lasting security is closely related to everlasting security [61]. Everlasting security requires secure quantum channels whereas long-lasting secure protocols can be deployed using existing post-quantum infrastructure, hence, long-lasting is more realistic security notion. Our long-lasting secure voting protocol provides vote privacy against future quantum adversaries. We provide a detailed comparison with other relevant voting protocols in Appendix. A.

Protocols	Non-Interactive Dealer	Long-lasting Security	Post-Quantum Security	Assumptions
Discrete-Log-based+Multisignatures [36, 50]	×	×	×	Discrete Log+Multisignatures
Discrete-Log-based [26, 33, 55]	✓	×	×	Discrete Log
Class-Group-based [24, 43]	✓	×	×	q-Hard subgroup membership
Signature-based [27]	×	✓	×	Perfectly hiding PCS+Signatures+ PQ-PKE (LWE/RLWE)
Bivariate Polynomial-based+ Multisignatures [5, 7]	✓	✓	×	Perfectly hiding PCS+Multisignatures+ PQ-PKE (LWE/RLWE)
Lattice-based [4, 49]	✓	✓	✓	LWE/Ring-LWE
Lattice+DL-based [37]	✓	✓	×	LWE/Ring-LWE+Discrete Log
Hash-based [10, 13]	×	✓	✓	RO + PQ-PKE (LWE/RLWE)
Ours (Hash-based)	✓	✓	✓	RO + PQ-IBE (LWE/RLWE)
Ours (Commitment-based)	✓	✓	×†	AHC + PQ-IBE (LWE/RLWE)
Note: PQ-PKE: Post-quantum Secure Public Key encryption, RO: Random Oracle, AHC: Perfectly Hiding Additively Homomorphic Commitment † Our Commitment-based PVSS protocol has aggregatable transcripts that enabling different applications. The hash-based PVSS lacks that property.				

Table 1: Comparison of Publicly Verifiable VSS Schemes.

1.3 Technical Overview

A popular line of PVSS protocols with non-interactive dealers follow the commit–encrypt–prove paradigm: the dealer commits to a polynomial $f(x)$ with secret $s := f(0)$, encrypts each share $[\![s]\!]_i := f(i)$ under the receivers' public keys, and provides a NIZK proof that each encrypted $f(i)$ is an evaluation of the committed polynomial $f(x)$. Classical DL-based approaches [26, 33, 36, 50, 55] yield highly efficient schemes using group homomorphisms but are vulnerable to quantum attacks via Shor's algorithm [57]. Lattice-based PVSS schemes [4, 49] in this paradigm offer post-quantum security but rely on costly lattice-based NIZKs, making them computationally expensive.

Another line of work [5, 7, 47] explores dealing the secret using a bivariate polynomial so that the receivers error-correct the shares without dealer interaction. However, this approach incurs quadratic communication overhead for the dealer.

We explore a third paradigm [54] where the dealer neither incurs the computation overhead of an NIZK nor does it incur the quadratic communication overhead of sharing a bivariate polynomial. We observe that in the commit-encrypt-prove paradigm the NIZK proof can be broken down into two parts:

(i) The commitment to secret s and the commitment to the shares $[\![s]\!]_i$ lie on a committed t-degree polynomial, and

(ii) The committed shares are indeed the same as the encrypted values under each receiver's public key.

The first statement can be performed quite efficiently, e.g. using a non-interactive threshold ZK proofs [10] if the commitment is non-homomorphic, or using a SCRAPE test [23] if the commitment is additively homomorphic. The second statement is expensive to prove because the NIZK uses the encryption and commitment in a non-black-box fashion. Therefore, one key idea in our approach is to defer this verification to a later phase, which does not involve the dealer, and yet achieves public verifiability.

One-Time Public Key Encryption Approach: The high-level intuition is that the dealer encrypts the share and the decommitment and posts these encryptions and commitments on a public bulletin board or a blockchain, denoted by $\mathcal{F}_{\mathsf{BC}}$. The receivers read the commitments on $\mathcal{F}_{\mathsf{BC}}$ and check that the committed shares indeed lie on a t-degree polynomial. Then, each i-th receiver decrypts its share (and opening randomness) from the i-th encryption. If the decrypted values is not a correct opening to the i-th committed share, then the i-th receiver simply reveals their secret key on $\mathcal{F}_{\mathsf{BC}}$. Anyone can publicly decrypt the dealer-posted i-th encryption using the this decryption key, and verify the validity of the i-th receiver's claim. This avoids using NIZK for step-(ii): encrypted share and the committed share is the same.

Non-Interactive Key Exchange Approach: Leaking the receiver's long-term secret key is non-scalable, requiring fresh public-key setup for each incident and preventing key reuse across applications. Prior work like Eth-DKG [54] uses Diffie-Hellman key exchange: the dealer and each receiver maintain DH public keys, enabling pairwise secret key establishment for encrypting shares and opening randomness. For complaints, receivers reveal the pairwise secret and provide efficient Chaum-Pedersen [25] NIZK proofs of correctness. Public complaint verification involves checking the NIZK proof, decrypting the share/randomness, and validating against the commitment. This approach is limited to discrete-log settings due to cheap NIZK proofs. Extension to lattice-based key exchange [34] is prohibitively expensive: efficient NIZK proofs for the required relations are unknown, and millisecond-level lattice-based decryption proofs remain unavailable.

Our Protocol: We circumvent the need for a lattice-based NIZK by replacing the public key encryption with an IBE [6, 32] – we use a lattice-based construction to obtain post-quantum security. In a one-time setup phase, each i-th receiver

publishes an individual master IBE public key mpk_i for which they store a master secret key msk_i. The dealer commits to the shares and the secret, and encrypts the openings to the i-th shares under each mpk_i and the dealer's identity id_d. The dealer posts these commitments and encryptions, and then goes offline. The receivers check that the posted commitments commit to a degree t sharing of a polynomial. Then, each i-th receiver checks that the i-th encrypted values is indeed a valid decommitment to the i-th commitment – if not, then the receiver uses msk_i to extract dealer's decryption key $\mathsf{sk}_{i,\mathsf{id}_d}$ and post it on $\mathcal{F}_{\mathsf{BC}}$. It takes $<$ 5ms to generate identity secret key. Anyone (even someone who is neither the dealer nor a receiver) can decrypt the i-th ciphertext using $\mathsf{sk}_{i,\mathsf{id}_d}$ and verify whether the receiver's claim is correct or not by matching the decrypted values with the commitment – this ensures public verifiability. Decrypting the IBE ciphertext takes $< 1ms$. This approach yields two orders of magnitude improvement in overall runtime, while maintaining crucial properties such as public verifiability and non-interactive dealers. Concretely, our dealer deals a secret among 1000 receivers in $<$ 1 sec.

(Informal) Security Argument: Clearly, the only way a dealer deals a bad share without getting detected is if it breaks the binding of the commitment scheme, otherwise it always gets caught. Meanwhile, the dealer's shares remain private due to the hiding of the commitment and the CPA-security of the encryption. We note that a malicious receiver cannot frame an honest dealer as corrupt by producing a bad secret key $\mathsf{sk}'_{i,\mathsf{id}_d}$ s.t. a valid ciphertext under $(\mathsf{mpk}_i, \mathsf{id}_d)$ decrypts to something else. This follows from the newly defined key verifiability property of the IBE scheme, which enforces that an adversarial receiver cannot produce a master public key mpk_i, s.t. it can later release a decryption keys$\mathsf{sk}'_{i,\mathsf{id}_d}$ which verifies correctly, but leads to an incorrect decryption.

Concrete Instantiations: To complete our protocol, we need a PVSS scheme with a non-interactive dealer that allows verifying that the committed shares correspond to a t-degree sharing of a polynomial. Our post-quantum secure instantiation consists of the hash-based VSS protocol of [10]. In their PVSS, once the dealing is done, the dealer uses a distributed ZK proof (DZK) to prove that the committed shares lie on a t-degree polynomial, and in case, there is a complaint by a receiver the dealer broadcasts the opening to the receiver's share. We note that it suffices for us that the committed shares are partially verified as the IBE allows a receiver to prove later on that the dealer dealt bad shares, while removing the dealer interaction during the share verification phase. The DZK proofs are instantiated using hashes, modeled as a random oracle.

For our long-lasting secure instantiation, we use a variant of Pedersen PVSS [50] where the dealer commits to the secret and the shares using (homomorphic) Pedersen commitments. Due to homomorphism, the degree-check (part-(i) of the proof) is possible using a so-called SCRAPE test (described in Appendix. B.1). The rest of the protocol stays unaltered: the receiver outputs the identity secret key in case the encrypted share does not match the committed share. Finally, Pedersen's unconditional hiding along with the post-quantum CPA-security of the IBE scheme yield long-lasting security by ensuring the post-quantum privacy of the secret.

2 Preliminaries

Notations We use $\mathbb{N}$ to denote a set of positive integers, and $[n]$ to denote the set $\{1,\ldots,n\}$ for any $n \in \mathbb{N}$. We denote the security parameter by $\lambda \in \mathbb{N}$. The symbol $\perp$ denotes an undefined value, or invalidity. $\mathsf{negl}(\lambda)$ is a negligible function in security parameter λ. A set $X = \{x_1,\ldots,x_n\}$ is denoted as $x_{[n]}$ or $\{x_i\}_{i\in[n]}$. For any subset $S \subset [n]$, x_S or $\{x_i\}_{i\in S}$ denotes the subset of X containing all x_i's such that $i \in S$. A ordered tuple $(x_1,\ldots,x_n)$ is denoted by vector notation $\vec{x}_{[n]}$ or $(x_i)_{i\in[n]}$. Similarly for any subset S of $[n]$, $\vec{x}_S$ and $(x_i)_{i\in S}$ are defined. Every algorithm takes the computational security parameter λ and statistical security parameter μ as inputs. We use $y := D(x)$ to denote a deterministic computation and $y := x$ for assignment. Randomized computations are denoted as $y \leftarrow R(x)$ by running a randomized algorithm $R(\cdot)$ on input x. We discuss the SCRAPE test from [23] in Appendix. B.1. We denote indistinguishability against a PPT quantum adversary using $\overset{\mathsf{pq}}{\approx}$ and indistinguishability against a PPT classical adversary using $\overset{\mathsf{c}}{\approx}$ respectively.

Communication Channels We assume that all the parties have read and write access to a public bulletin board or a blockchain [28], denoted as $\mathcal{F}_{\mathsf{BC}}$. For secure voting we additionally need smart contracts which can be formally modeled using [39]. For point-to-point communication we assume secure, authenticated channels between the parties.

Commitment Schemes We denote a perfectly hiding and computationally binding commitment Com to message $m \in \mathcal{M}$ under randomness $r \in \mathcal{R}$ as $\langle m; r\rangle := \mathsf{Com}(m; r)$. We assume that Com is additively homomorphic, i.e. $\langle m_1; r_1\rangle \cdot \langle m_2; r_2\rangle = \langle m_1 + m_2; r_1 + r_2\rangle$ for all $m_1, m_2 \in \mathcal{M}$ and all $r_1, r_2 \in \mathcal{R}$. For concrete instantiation, we assume Com is Pedersen Commitment [50].

2.1 Identity-Based Encryption

An Identity-Based Encryption (IBE) scheme [17] consists of the following probabilistic polynomial-time algorithms. We also define an additional algorithm VerKey to verify a secret key $\mathsf{sk}_{\mathsf{ID}}$ corresponding to an identity ID.

- $\mathsf{KeyGen}(1^\lambda) \to (\mathsf{mpk}, \mathsf{msk})$: Outputs a master public key mpk and a master secret key msk.
- $\mathsf{Extract}(\mathsf{msk}, \mathsf{ID}) \to \mathsf{sk}_{\mathsf{ID}}$: Outputs the private key $\mathsf{sk}_{\mathsf{ID}}$ corresponding to identity ID.

- $\mathsf{Enc}(\mathsf{mpk}, \mathsf{ID}, m) \to c$: Encrypts the message m as ciphertext c under master public key mpk and identity ID.
- $\mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}}, c) \to m$: Decrypts c to obtain message m.
- $\mathsf{VerKey}(\mathsf{mpk}, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}}) \to \{0,1\}$: Outputs 1 if $\mathsf{sk}_{\mathsf{ID}}$ is a valid secret key corresponding to ID, else outputs 0.

The IBE scheme needs to satisfy correctness and IND-ID-CPA Security [17]. In addition, we need a new property called key verifiability that allows checking whether a generated secret key $\mathsf{sk}_{\mathsf{ID}}$ is valid or not w.r.t. mpk and ID. We define it as.

Key Verifiability: An IBE scheme is said to satisfy key verifiability if for any security parameter λ, any identity ID, any message m, and any stateful PPT adversary $\mathcal{A}$, the following probability is negligible in λ.

$$\Pr\left[\begin{array}{c|c} \mathsf{VerKey}(\mathsf{mpk}, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}}) = 1; & \mathsf{mpk} \leftarrow \mathcal{A}(1^\lambda); \\ m \neq \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}}, c) & c \leftarrow \mathsf{Enc}(\mathsf{mpk}, \mathsf{ID}, m); \\ & \mathsf{sk}_{\mathsf{ID}} \leftarrow \mathcal{A}(c) \end{array}\right]$$

Informally, except with negligible probability, an adversary can not produce a master public key, such that it can later release a decryption key which verifies correctly, but leads to an incorrect decryption. The Boneh-Franklin IBE scheme [17] and the lattice-based IBE scheme [32] satisfy key verifiability.

Post Quantum Secure IBE: Finally, we say that an IBE scheme is post-quantum secure if all of the above properties hold against an adversary with quantum capabilities.

We use the lattice-based IBE scheme of [32] for our protocol. It is based on the GPV scheme [38] over NTRU lattices and it is post-quantum secure under the Ring-LWE [48] assumption. In the security proofs of our PVSS protocols, we also argue the adaptive security of the schemes. For that, we need the IBE encryptions to be non-committing [22], i.e., a simulated ciphertext can be later opened consistently as an encryption of any message. We note that the IBE in [32] is non-committing in the programmable random oracle model. We refer to Appendix. B.3 for more details.

2.2 Shamir's Secret Sharing Scheme

We recall the definition of an (n,t) Shamir secret sharing scheme [56] as follows:

Definition 2.1 (Shamir's Secret Sharing Scheme)**.** Let $\mathbb{F}$ be a finite field. It consists of two algorithms:

- $\mathsf{Share}(s,n,t) \to \{[\![s]\!]_i\}_{i\in[n]}$: Shares a secret s using a random t-degree polynomial to generate n shares $\{[\![s]\!]_i\}_{i\in[n]}$.
- $\mathsf{Recon}(z, \{i_j, [\![s]\!]_{i_j}\}_{j=1}^{t+1}) \to f(z)$: Given any $t+1$ shares, it outputs evaluation value $f(z)$ at point z.
- $\mathsf{ReconPoly}(\{i_j, [\![s]\!]_{i_j}\}_{j=1}^{t+1}) \to f(x)$: Given any $t+1$ shares, it outputs the polynomial $f(x)$.

Shamir sharing scheme is also defined over an additively homomorphic commitment scheme defined over message space $\mathcal{M} = \mathbb{F}$ and randomness space $\mathcal{R} = \mathbb{F}$. It consists of the following additional algorithms:

- $\mathsf{ShareCom}(n,t) \to (s, r, \{[\![s]\!]_i, [\![r]\!]_i, \langle [\![s]\!]_i; [\![r]\!]_i \rangle\}_{i\in[n]})$: Shares a random secret s and a randomness r using two t-degree polynomials to generate n shares $\{[\![s]\!]_i\}_{i\in[n]}$, $\{[\![r]\!]_i\}_{i\in[n]}$ and commitments to those shares as $\{\langle [\![s]\!]_i; [\![r]\!]_i \rangle\}_{i\in[n]}$.
- $\mathsf{ShareCom}(s,n,t) \to (r, \{[\![s]\!]_i, [\![r]\!]_i, \langle [\![s]\!]_i; [\![r]\!]_i \rangle\}_{i\in[n]})$: Shares secret s and a randomness r using two t-degree polynomials to generate n shares $\{[\![s]\!]_i\}_{i\in[n]}$, $\{[\![r]\!]_i\}_{i\in[n]}$ and commitments to those shares as $\{\langle [\![s]\!]_i; [\![r]\!]_i \rangle\}_{i\in[n]}$.
- $\mathsf{ReconCom}(z, \{i_j, \langle [\![s]\!]_{i_j}; [\![r]\!]_{i_j} \rangle\}_{j=1}^{t+1}) \to \langle s; r \rangle$: Given any $t+1$ shares, it reconstructs the committed secret $\langle s; r \rangle$.

The above scheme satisfies correctness and perfect secrecy. Full details in Appendix. B.4.

3 Security Model and Security Definitions

3.1 Definition: Publicly Verifiable Secret Sharing with Non-interactive Dealer

A (n,t) *Verifiable Secret Sharing (VSS)* [10, 27] consists of the following PPT algorithms:

1. $\mathsf{Setup}(1^\lambda) \to (\mathsf{pp}, \{\mathsf{pk}_i, \mathsf{sk}_i\}_{i\in[n]})$: Given the security parameter λ as input, the public parameters are generated, and each party generates its own public and secret key pair. The public parameters and the public keys are shared among all the parties and they are implicit inputs to all the following algorithms.
2. $\mathsf{Share}(s,n,t) \to (\{[\![s]\!]_i, e_i\}_{i\in[n]}, \pi)$: Given the number of parties n, corruption threshold t, and the secret s, the dealer computes shares $[\![s]\!]_i$ for the i-th party and the corresponding encryption under pk_i as e_i. The dealer also computes a proof π of correct sharing. The dealer outputs $(\{[\![s]\!]_i, e_i\}_{i\in[n]}, \pi)$.
3. $\mathsf{Decrypt}(e_i, \mathsf{sk}_i) \to ([\![s]\!]_i, \mathsf{st}_i)$: Given the encrypted share e_i and secret key sk_i compute the decrypted share as $[\![s]\!]_i$ and a publicly verifiable decryption state st_i $(\in \{0,1\}^*)$.
4. $\mathsf{PubVer}(\mathsf{st}_i, e_i, \pi, n, t) \to \{0,1\}$: Given the public decryption state st_i any party can verify that the encrypted share e_i is invalid or not, corresponding to the proof π. If the verification succeeds, then PubVer outputs 1, else 0.
5. $\mathsf{Recon}(\{i_j, [\![s]\!]_{i_j}\}_{j\in[t+1]}, \pi, n, t) \to s/\mathcal{L}$: Given any $t+1$ shares, the algorithm either reconstructs the secret s or outputs a list $\mathcal{L}$ of malformed shares.

Public Verifiability: A VSS is publicly verifiable if the transcript of the VSS protocol can be publicly verified by any external party to check that the dealing was valid/invalid.

Given the decryption state st_i any external party can publicly verify, via PubVer algorithm, whether the encrypted share e_i is valid or not, given the proof π.

Definition 3.1 (Computationally Secure PVSS). A publicly verifiable VSS scheme (PVSS) is computationally secure if the dealer and receiver and all adversaries are computationally bounded algorithms (PPT) and the scheme satisfies the following properties:

- *Correctness:* Any set of $t+1$ or more valid shares correctly reconstructs the secret s.
- *Termination:* All honest nodes will eventually terminate the Share, Decrypt and PubVer phase.
- *Reconstructability:* If a dealing verifies successfully, then, given any set of $\geq t+1$ shares, the Recon algorithm either reconstructs a unique s or identifies a non-empty list $\mathcal{L} \neq \emptyset$ of malformed shares even if the dealer is adversarial.
- *Computational Secrecy:* Assuming an honest dealer dealing secret s, then any set of $\leq t$ shares, $\leq t$ secret keys, n states $\{\mathsf{st}_i\}_{i\in[n]}$, n encrypted shares and π hides s against an adversary, i.e. for any two secrets s, s' following holds:

$$\{s, \{i_j, [\![s]\!]_{i_j}, \mathsf{sk}_i\}_{j\in[t]}, \{e_i, \mathsf{pk}_i\}_{i\in[n]}, \pi\} \overset{c}{\approx}$$

$$\{s', \{i_j, [\![s]\!]_{i_j}, \mathsf{sk}_i\}_{j\in[t]}, \{e_i, \mathsf{pk}_i\}_{i\in[n]}, \pi\},$$

  where the $\{i_j\}_{j\in[t]}$ indices are chosen by the adversary.

Post-Quantum Secure PVSS: We say that a PVSS scheme achieves post-quantum security if all the properties hold even against quantum PPT adversaries.

Long-lasting Secure PVSS: We say that a PVSS scheme achieves long-lasting security if all properties except computational secrecy holds against classical PPT adversaries, but the secrecy holds against quantum PPT adversaries.

Non-interactive Dealers: A PVSS is said to have non-interactive dealers if the dealer non-interactively performs the sharing by locally producing its message and sending it over and then each receiver can verify that their share $[\![s]\!]_i$, obtained via Decrypt, is valid without further interaction with the dealer.

3.2 A New Model: Long-lasting Security

A protocol is said to be long-lasting secure if it is secure against a PPT classical adversary *during* the protocol execution. And after the protocol terminates, the protocol transcript would remain secure (does not leak any information about honest parties' inputs) against a future PPT quantum adversary. We formally define the notion of long-lasting security following the simulation-based real-ideal world paradigm of [21] in Appendix. C, Definition. C.1. We note that our PVSS protocol satisfies game-based definition and so we do not need the simulation-based definition for this paper. However, we believe this model can be interesting independently for building schemes that are to be deployed today, but require long-term privacy.

4 Our Post-Quantum Secure PVSS Scheme

We build our PVSS scheme on top of the distributed ZK proof and hash-based PVSS of [10]. In the PVSS protocol of [10], the dealer shares its secret s and commits to it and its shares using (hash-based) non-homomorphic commitments. Then, the dealer computes a distributed ZK proof (DZK) using a Schnorr-like proof on committed polynomials to prove that the sharing lies on a t-degree polynomial. The dealer sends the shares and the proofs to each receiver. Upon receiving its share and part of the ZK proof, each receiver verifies its local share, and raises a complaint if its ZK proof does not verify. In case of a complaint, then the dealer leaks the receiver's share, and anyone can verify the legitimacy of the claim. Their PVSS protocol requires the dealer to interact to resolve the receivers' complaints. The overview can be found in Section 1.3.

In our PVSS, the dealer encrypts the receiver's state under the receiver's master public key and the dealer's identity. To raise a complaint, the receiver produces the identity secret key, and anyone can decrypt the receiver's state(encrypted by the dealer) and verify the legitimacy of the claim. This removes the dealer interaction from the PVSS protocol.

We present the formal protocol details below. We assume that (KeyGen, Extract, Enc, Dec) is a post-quantum secure IBE scheme, H is a random oracle, and (Share, Recon, ReconPoly) is a Shamir secret sharing scheme.

Setup(1^λ):

- For $i \in [n]$, each i-th receiver runs $(\mathsf{mpk}_i, \mathsf{msk}_i) \leftarrow \mathsf{KeyGen}(1^\lambda)$ to obtain master public-secret keys.
- All the n master public keys - $(\mathsf{mpk}_1, \ldots, \mathsf{mpk}_n)$ are posted on the bulletin board $\mathcal{F}_{\mathsf{BC}}$.

Share($s \in \mathbb{F}, n, t$): The dealer performs the following:

1. *Dealing:* Generates shares of s using t-degree polynomial $f(x)$ s.t. $f(0) := s$. The dealer also shares a commitment randomness r.

   $$\{[\![s]\!]_i\}_{i\in[n]} := \mathsf{Share}(s,n,t), \{[\![r]\!]_i\}_{i\in[n]} := \mathsf{Share}(n,t)$$

   Commit the share as $\mathsf{csh}_i := \mathrm{H}([\![s]\!]_i, [\![r]\!]_i)$ for $i \in [n]$.
2. *First message of Distributed-ZK (DZK):* Sample a random t-degree padding polynomial $B(x)$ with coefficients in the field $\mathbb{F}$. For each $i \in [n]$: sample randomness $q_i \leftarrow \{0,1\}^\lambda$. Compute commitments

to $B(i)$ as

$$\mathsf{cpt}_i := \mathrm{H}(B(i), q_i) \text{ for } i \in [n].$$

3. *Challenge of DZK:* Set $\mathsf{ch} := \mathrm{H}(\{\mathsf{csh}_i, \mathsf{cpt}_i\}_{i\in[n]})$.
4. *Opening of DZK:* Set $\mathsf{c} := W(x) := B(x) - \mathsf{ch} \cdot f(x)$.
5. *Encrypting Receivers' States:* For $i \in [n]$, encrypt the i-th receiver's state under $\mathsf{ID}_i := (\mathsf{Dealer}, i)$ as:

$$\mathsf{E}_i = \mathsf{Enc}(\mathsf{mpk}_i, \mathsf{ID}_i, ([\![s]\!]_i, [\![r]\!]_i, q_i)).$$

6. Set PVSS proof $\pi := (\mathsf{c}, \{\mathsf{csh}_i, \mathsf{cpt}_i\}_{i\in[n]})$. Posts encryptions $\{\mathsf{E}_i\}_{i\in[n]}$ and proof π on $\mathcal{F}_{\mathsf{BC}}$.

$\underline{\mathsf{Decrypt}(\mathsf{E}_i, \mathsf{msk}_i)}$: Each i-th receiver runs the following:

1. Decrypts its shares and randomness values as $([\![s]\!]_i, [\![r]\!]_i, q_i) := \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}_i}, \mathsf{E}_i)$ where $\mathsf{sk}_{\mathsf{ID}_i} := \mathsf{Extract}(\mathsf{msk}_i, (\mathsf{Dealer}, i))$.
2. Set $\mathsf{st}_i := (\textsc{Complaint}, i, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}_i})$ if
   - The committed share does not decommit correctly, i.e. $\mathrm{H}([\![s]\!]_i, [\![r]\!]_i) \neq \mathsf{csh}_i$, OR
   - The DZK failed to verify for i-th receiver, i.e. $\mathrm{H}(W(i) + \mathsf{ch} \cdot [\![s]\!]_i, q_i) \neq \mathsf{cpt}_i$ where $\mathsf{ch} := \mathrm{H}(\{\mathsf{csh}_i, \mathsf{cpt}_i\}_{i\in[n]})$.

   Otherwise, set $\mathsf{st}_i := \emptyset$.
3. Post st_i on $\mathcal{F}_{\mathsf{BC}}$, and output $([\![s]\!]_i, [\![r]\!]_i, q_i)$ and st_i.

$\underline{\mathsf{PubVer}(\mathsf{st}_i, \mathsf{E}_i, \pi, n, t)}$: Any party can run this as follows:

1. Parse PVSS proof $(\mathsf{c}, \{\mathsf{csh}_i, \mathsf{cpt}_i\}_{i\in[n]}) := \pi$.
2. If $W(x) := \mathsf{c}$ is not a t-degree polynomial then output 0 and discard the dealings.
3. If there are no complaints, i.e. $\mathsf{st}_i == \emptyset$ for all $i \in [n]$, then output 1. Else, continue to the next step.
4. For all complaints (posted on $\mathcal{F}_{\mathsf{BC}}$) of the form $(\textsc{Complaint}, i, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}_i})$: Reject the complaint if

$$(\mathsf{VerKey}(\mathsf{mpk}_i, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}_i}) == 0) \vee (\mathrm{H}([\![s]\!]_i, [\![r]\!]_i) == \mathsf{csh}_i$$
$$\& \ \mathrm{H}(W(i) + \mathsf{ch} \cdot [\![s]\!]_i, q_i) == \mathsf{cpt}_i),$$

   where $([\![s]\!]_i, [\![r]\!]_i, q_i) := \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}_i}, \mathsf{E}_i)$ and $\mathsf{ch} := \mathrm{H}(\{\mathsf{csh}_i, \mathsf{cpt}_i\}_{i\in[n]})$. Output 0 if the complaint is valid.

If $W(x)$ is an t-degree polynomial and none of the raised complaints were valid, then output 1.

$\underline{\mathsf{Recon}(\{i_j, [\![s]\!]_{i_j}, [\![r]\!]_{i_j}, q_{i_j}\}_{j\in[t+1]}, \pi, n, t)}$:

1. Initialize list $\mathcal{L} := \emptyset$ for storing malformed shares.
2. *Output Share Verification:* Given any share $\{i_j, [\![s]\!]_{i_j}, [\![r]\!]_{i_j}, q_{i_j}\}_{j\in[t+1]}$ if $\mathsf{cpt}_{i_j} \neq \mathrm{H}(W(i_j) + \mathsf{ch} \cdot [\![s]\!]_{i_j}, q_{i_j}) \vee \mathsf{csh}_{i_j} \neq \mathrm{H}([\![s]\!]_{i_j}, [\![r]\!]_{i_j})$, where $\mathsf{ch} := \mathrm{H}(\{\mathsf{csh}_i, \mathsf{cpt}_i\}_{i\in[n]})$, then consider it as invalid share and add i_jth share to discard list as $\mathcal{L} := \mathcal{L} \cup \{i_j\}$.
3. Given $t+1$ valid shares that pass the above check, reconstruct $f(x)$ and $r(x)$ polynomials as

$$f(x) := \mathsf{ReconPoly}(\{i_j, [\![s]\!]_{i_j}\}_{j\in[t+1]}),$$
$$r(x) := \mathsf{ReconPoly}(\{i_j, [\![r]\!]_{i_j}\}_{j\in[t+1]}).$$

   Output $s := f(0)$ as the secret, $f(x)$ as the secret polynomial and $r(x)$ as the randomness polynomial corresponding to the commitments $\mathsf{csh}_i = \mathrm{H}(f(i), r(i))$ for $i \in [n]$.

We prove that the above PVSS protocol is post-quantum secure by proving Thm. 4.1 in Appendix. D.1, and we also briefly discuss adaptive security of our PVSS protocol. We note that our protocol is secure up to $t < n$ corruptions, but for liveness we need $t < \frac{n}{2}$.

Theorem 4.1. *Assuming that* $(\mathsf{KeyGen}, \mathsf{Extract}, \mathsf{Enc}, \mathsf{Dec})$ *is a post quantum secure IND-ID-Secure IBE scheme with key verifiability and* $(\mathsf{Share}, \mathsf{Recon}, \mathsf{ReconPoly})$ *is a Shamir secret sharing scheme (Sec.2.2), then the above protocol is a post-quantum secure PVSS (Sec. 3.1) with non-interactive dealer and n receivers against an adversary corrupting upto* $t < \frac{n}{2}$ *parties in the random oracle model.*

5 Long-lasting Secure PVSS Construction and Polling

Our post-quantum secure PVSS protocol did not provide an aggregatable transcript since the commitments csh and cpt are instantiated using hash-based PQ-secure commitments. This limits its usage in different aggregation protocols like secure voting, statistics computation, and so on, where we need these commitments to be aggregated. One could consider replacing the hashes with an additively homomorphic commitment, but that would require $2n$ commitments in proof π. We instead reduce that to n commitments using a more optimized PVSS protocol that has aggregatable transcripts. For that, we relax the security to longlasting as it suffices for our longlasting secure voting protocol.

5.1 Longlasting Secure PVSS

Our longlasting PVSS protocol uses a computationally binding and perfectly hiding additively homomorphic commitment scheme Com, instantiated using Pedersen Commitments. In addition, it uses the IBE scheme and a Shamir sharing scheme - $(\mathsf{Share}, \mathsf{Recon}, \mathsf{ShareCom}, \mathsf{ReconCom})$, defined (in Sec. 2.2) over commitment Com. Our protocol uses

a SCRAPE test [23] over committed shares to publicly check that the underlying committed polynomial is t-degree. The previous PVSS protocol did this using the distributed ZK proof. We provide the details of the SCRAPE test (denoted as $\mathcal{F}_{\mathsf{SCRAPE}}$) in Appendix. B.1 for completeness. The high-level overview of our PVSS can be found in Sec. 1.3, and the protocol details are as follows.

$\mathsf{Setup}(1^\lambda)$:

- For $i \in [n]$, each i-th receiver runs $(\mathsf{mpk}_i, \mathsf{msk}_i) \leftarrow \mathsf{KeyGen}(1^\lambda)$ to obtain master public-secret keys.
- All the n master public keys - $(\mathsf{mpk}_1, \ldots, \mathsf{mpk}_n)$ are posted on the bulletin board $\mathcal{F}_{\mathsf{BC}}$.

$\mathsf{Share}(s,n,t)$: The dealer performs the following:

1. *Dealing:* Generates committed shares of secret s using randomness r.

   $$(s, r, \{[\![s]\!]_i, [\![r]\!]_i, \langle [\![s]\!]_i; [\![r]\!]_i \rangle\}_{i\in[n]}) := \mathsf{ShareCom}(s,n,t)$$

   For $i \in [n]$, denote the committed shares as

   $$\mathsf{csh}_i := \langle [\![s]\!]_i, [\![r]\!]_i \rangle.$$

   Denote the committed secret as $\mathsf{c} := \langle s; r \rangle$.

2. *Encrypting Receivers' States:* For $i \in [n]$, encrypt i-th receiver's share under $\mathsf{ID}_i := (\mathsf{Dealer}, i)$ as E_i:

   $$\mathsf{E}_i = \mathsf{Enc}(\mathsf{mpk}_i, \mathsf{ID}_i, ([\![s]\!]_i, [\![r]\!]_i)).$$

3. Sets PVSS proof $\pi := \{\mathsf{c}, \{\mathsf{csh}_i\}_{i\in[n]}\}$.
4. The dealer posts encryptions $\{\mathsf{E}_i\}_{i\in[n]}$ and proof π on $\mathcal{F}_{\mathsf{BC}}$.

$\mathsf{Decrypt}(\mathsf{E}_i, \mathsf{msk}_i)$: Each i-th receiver runs the following:

1. Decrypt shares as $([\![s]\!]_i, [\![r]\!]_i) := \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}_i}, \mathsf{E}_i)$ where $\mathsf{sk}_{\mathsf{ID}_i} := \mathsf{Extract}(\mathsf{msk}_i, (\mathsf{Dealer}, i))$.
2. Set $\mathsf{st}_i := (\textsc{Complaint}, i, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}_i})$ if $\mathsf{Com}([\![s]\!]_i; [\![r]\!]_i) \neq \mathsf{csh}_i$. Otherwise, set $\mathsf{st}_i := \emptyset$.
3. Post st_i on $\mathcal{F}_{\mathsf{BC}}$, and output $([\![s]\!]_i, [\![r]\!]_i)$ and st_i.

$\mathsf{PubVer}(\mathsf{st}_i, \mathsf{E}_i, \pi, n, t)$: Any party can run this as:

1. Parse PVSS proof as $\{\mathsf{c}, \{\mathsf{csh}_i\}_{i\in[n]}\} := \pi$.
2. If $\mathcal{F}_{\mathsf{SCRAPE}}((n,t), \mathsf{c}, \mathsf{csh}_1, \ldots, \mathsf{csh}_n) == 0$ then output 0 and discard the dealings.
3. If there are no complaints, i.e. $\mathsf{st}_i == \emptyset$ for all $i \in [n]$, then output 1. Else, continue to the next step.
4. For all complaints (posted on $\mathcal{F}_{\mathsf{BC}}$) of the form $(i, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}_i})$: Reject the complaint if

   $$(\mathsf{VerKey}(\mathsf{mpk}_i, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}_i}) == 0) \vee$$

   $$(\mathsf{Com}([\![s]\!]_i; [\![r]\!]_i) == \mathsf{csh}_i),$$

   where $([\![s]\!]_i, [\![r]\!]_i) := \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}_i}, \mathsf{E}_i)$. If the complaint is valid, then output 0.

If the SCRAPE test passed and none of the raised complaints were valid then output 1.

$\mathsf{Recon}(\{i_j, [\![s]\!]_{i_j}, [\![r]\!]_{i_j}\}_{j\in[t+1]}, \pi, n, t)$:

1. Initialize list $\mathcal{L} := \emptyset$ for storing malformed shares.
2. *Output Share Verification:* Given any $t+1$ shares of the form $\{i_j, [\![s]\!]_{i_j}, [\![r]\!]_{i_j}\}_{j\in[t+1]}$ if $\mathsf{csh}_{i_j} \neq \mathsf{Com}([\![s]\!]_{i_j}; [\![r]\!]_{i_j})$ then consider it as invalid share and add i_jth share to discard list as $\mathcal{L} := \mathcal{L} \cup \{i_j\}$.
3. Given $t+1$ valid shares that pass the above check, reconstruct the secret s and randomness r as

   $$s := \mathsf{Recon}(0, \{i_j, [\![s]\!]_{i_j}\}_{j\in[t+1]}),$$

   $$r := \mathsf{Recon}(0, \{i_j, [\![r]\!]_{i_j}\}_{j\in[t+1]}).$$

   Output s as the secret and r as the randomness corresponding to commitment $\mathsf{c} = \mathsf{Com}(s; r)$.

We prove that the above PVSS protocol is longlasting secure by proving Thm. 5.1 in Appendix. D.2, and we also briefly discuss adaptive security of our PVSS protocol. We note that our protocol is secure up to $t < n$ corruptions, but for liveness we need $t < \frac{n}{2}$.

Theorem 5.1. *Assuming that* Com *is a computationally binding and perfectly hiding additively homomorphic commitment scheme,* $(\mathsf{KeyGen}, \mathsf{Extract}, \mathsf{Enc}, \mathsf{Dec})$ *is a post quantum secure IND-ID-Secure IBE scheme with key verifiability and* $(\mathsf{Share}, \mathsf{Recon}, \mathsf{ShareCom}, \mathsf{ReconCom})$ *is a Shamir secret sharing scheme (defined in Sec. 2.2) over commitment* Com*, then the above protocol is a longlasting secure PVSS (Def. 3.1) with non-interactive dealer and n receivers against an adversary corrupting upto* $t < \frac{n}{2}$ *parties.*

We provide a proof sketch in Appendix. D.2 and we also discuss adaptive security of our PVSS protocol.

5.2 Longlasting Secure Polling

We present our polling protocol in this section. For polling, we use our longlasting PVSS protocol based on additive homomorphic commitments since the PVSS transcript is aggregatable and it allows us to generate additional proofs over the commitments very efficiently without sacrificing computational performance. The hash-based PVSS protocol

does not allow aggregation of the commitment c, and so it cannot be used in polling since c will consist of the committed votes and needs to satisfy additive homomorphism. We assume that Com is a computationally binding and perfectly hiding additively homomorphic commitment scheme, and $(\mathsf{Share}, \mathsf{Recon}, \mathsf{ShareCom}, \mathsf{ReconCom})$ is a Shamir secret sharing scheme defined over Com. We assume that all the parties have read and write access to $\mathcal{F}_{\mathsf{BC}}$. Our polling protocol consists of the following phases.

Client-Side Setup: Each client C is identified by its public verification key vk and possesses a corresponding secret key used to sign transactions. vk is posted on $\mathcal{F}_{\mathsf{BC}}$.

Server-Side Setup: There are n servers out of which $t < n$ of them can be corrupt. Each server S_i has an IBE master public key and master secret pair $(\mathsf{mpk}_i, \mathsf{msk}_i) \leftarrow \mathsf{Setup}(1^\lambda)$. All the n master public keys are posted on $\mathcal{F}_{\mathsf{BC}}$ as $\{\mathsf{S}_i, \mathsf{mpk}_i\}_{i\in[n]}$.

Setting up a Poll: To set up a poll with identifier pid, the organizer specifies the question, options (assume four for simplicity), and an expiry timestamp. The pid and expiry are posted on the smart contract, and once expiry is reached, servers compute the results.

Voting by a Client: Each poll client is allowed to vote for a single option in an active poll. To vote in a poll, identified as pid, the client performs the following:

1. Denote the vote vector as $(\mathsf{v}_1, \mathsf{v}_2, \mathsf{v}_3, \mathsf{v}_4)$ where $\mathsf{v}_j := 1$ if the client votes for jth option and $\mathsf{v}_{j'} := 0$ for all $j' \in [4] \setminus j$. For e.g., if the client votes for the second option then $(\mathsf{v}_1, \mathsf{v}_2, \mathsf{v}_3, \mathsf{v}_4) := (0, 1, 0, 0)$.
2. Compute the committed server shares of each value v_j, for $j \in [4]$, as:
   (a) The client generates committed shares of v_j as $(\mathsf{v}_j, \mathsf{r}_j, \{[\![\mathsf{v}_j]\!]_i, [\![\mathsf{r}_j]\!]_i, \langle [\![\mathsf{v}_j]\!]_i; [\![\mathsf{r}_j]\!]_i \rangle\}_{i\in[n]}) := \mathsf{ShareCom}(\mathsf{v}_j, n, t)$.
   (b) Denote the committed value as $\mathsf{c}_j := \langle \mathsf{v}_j, \mathsf{r}_j \rangle := \langle [\![\mathsf{v}_j]\!]_0, [\![\mathsf{r}_j]\!]_0 \rangle$.
   (c) Denote the committed shares as $\mathsf{csh}_{j,i} := \langle [\![\mathsf{v}_j]\!]_i; [\![\mathsf{r}_j]\!]_i \rangle$ for each server S_i.
3. For $i \in [n]$, encrypt the i-th server's shares under $\mathsf{ID}_i := (\mathsf{vk}, \mathsf{pid}, \mathsf{S}_i)$ as $\mathsf{E}_i := \mathsf{E}_i = \mathsf{Enc}(\mathsf{mpk}_i, \mathsf{ID}_i, \mathsf{msg}_i)$, where $\mathsf{msg}_i := ([\![\mathsf{v}_1]\!]_i, [\![\mathsf{r}_1]\!]_i, [\![\mathsf{v}_2]\!]_i, [\![\mathsf{r}_2]\!]_i, [\![\mathsf{v}_3]\!]_i, [\![\mathsf{r}_3]\!]_i, [\![\mathsf{v}_4]\!]_i, [\![\mathsf{r}_4]\!]_i)$.
4. Posts $(\mathsf{c}_1, \mathsf{c}_2, \mathsf{c}_3, \mathsf{c}_4, \{\mathsf{csh}_{1,i}, \mathsf{csh}_{2,i}, \mathsf{csh}_{3,i}, \mathsf{csh}_{4,i}, \mathsf{E}_i\}_{i\in[n]})$ on $\mathcal{F}_{\mathsf{BC}}$.
5. Proves that it has voted for a single option by proving that each committed vote is a bit, and they sum up to 1. The client computes the NIZK proof π_{vote} (Sec. E.1) as
$$\pi_{\mathsf{vote}} = \mathsf{Prove}_{\mathsf{vote}}((\mathsf{c}_1, \mathsf{c}_2, \mathsf{c}_3, \mathsf{c}_4, \mathsf{vk}, \mathsf{pid}), (\{\mathsf{v}_j, \mathsf{r}_j\}_{j\in[4]})),$$
$\mathsf{Prove}_{\mathsf{vote}}$ proves that $\forall j \in [4], (\mathsf{c}_j = \langle \mathsf{v}_j; \mathsf{r}_j \rangle)$ & $(\mathsf{c}_j = \mathsf{Com}(0) \vee \mathsf{c}_j = \mathsf{Com}(1))$ & $\mathsf{c}_1 + \mathsf{c}_2 + \mathsf{c}_3 + \mathsf{c}_4 = \mathsf{Com}(1)$.
6. Posts the following on $\mathcal{F}_{\mathsf{BC}}$ as a transaction and signs it using its secret key sk.
   (a) Verification key vk and poll identifier pid.
   (b) Commitments to the vote values as $(\mathsf{c}_1, \mathsf{c}_2, \mathsf{c}_3, \mathsf{c}_4)$,
   (c) Commitments to the servers' shares to the vote values as $\{\mathsf{csh}_{1,i}, \mathsf{csh}_{2,i}, \mathsf{csh}_{3,i}, \mathsf{csh}_{4,i}\}_{i\in[n]}$.
   (d) Proof π_{vote} of correct voting.
   (e) IBE encryptions to the servers' shares as $\{\mathsf{E}_i\}_{i\in[n]}$.

Note: The client's verification key vk and pid values are included in the proving algorithm for the NIZKs so that the clients encrypted votes cannot be replicated by another client and the vote is bound to $(\mathsf{vk}, \mathsf{pid})$.

Vote Verification by Smart Contract: The smart contract for pid verifies each client vote as follows:

- Verify that the client has not previously voted in pid by checking that there does exist a vote that starts with $(\mathsf{vk}, \mathsf{pid}, \ldots)$.
- Verify that the poll is active, by checking that the expiry timestamp of pid has not been reached.
- Verify that π_{vote} is valid (Sec. E.1) by running $\mathsf{Verify}_{\mathsf{vote}}((\mathsf{c}_1, \mathsf{c}_2, \mathsf{c}_3, \mathsf{c}_4, \mathsf{vk}, \mathsf{pid}), \pi_{\mathsf{vote}}) == 1$.

  If any of the above verification algorithms return 0, then mark this vote as invalid. Else, append the client's transaction on $\mathcal{F}_{\mathsf{BC}}$.

After a client vote is posted on the $\mathcal{F}_{\mathsf{BC}}$, the servers proceed to validate it and store the validated vote locally.

Client's Vote Validation: Each server S_i reads the client transaction, identified by $(\mathsf{vk}, \mathsf{pid}, \ldots)$, from $\mathcal{F}_{\mathsf{BC}}$ and then performs the following checks:

1. If any of the SCRAPE tests (Appendix. B.1) fail then discard the client, i.e. the following holds - $\exists j \in [4]$ s.t.
$$\mathcal{F}_{\mathsf{SCRAPE}}((n,t), \mathsf{c}_j, \mathsf{csh}_{j,1}, \ldots, \mathsf{csh}_{j,n}) = 0.$$
2. Extract the decryption key corresponding to $\mathsf{ID} := (\mathsf{vk}, \mathsf{pid}, \mathsf{S}_i)$ as $\mathsf{sk}_{\mathsf{ID}} := \mathsf{Extract}(\mathsf{msk}_i, \mathsf{ID})$.
3. Decrypt the server's shares as $\mathsf{msg}_i := \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}}, \mathsf{E}_i)$ from E_i and parse $([\![\mathsf{v}_1]\!]_i, [\![\mathsf{r}_1]\!]_i, [\![\mathsf{v}_2]\!]_i, [\![\mathsf{r}_2]\!]_i, [\![\mathsf{v}_3]\!]_i, [\![\mathsf{r}_3]\!]_i, [\![\mathsf{v}_4]\!]_i, [\![\mathsf{r}_4]\!]_i) := \mathsf{msg}_i$.
4. *Complaint Phase:* If any of the shares $[\![\mathsf{v}_j]\!]_i, [\![\mathsf{r}_j]\!]_i$ (for $j \in [4]$) fail to match with the commitment $\mathsf{csh}_{j,i}$, i.e. $\mathsf{csh}_{j,i} \neq \mathsf{Com}([\![\mathsf{v}_j]\!]_i; [\![\mathsf{r}_j]\!]_i)$ for some $j \in [4]$, then post a complaint on $\mathcal{F}_{\mathsf{BC}}$ as $(\mathsf{Complaint}, \mathsf{vk}, \mathsf{pid}, \mathsf{S}_i, j, \mathsf{sk}_{\mathsf{ID}})$ where $\mathsf{ID} := (\mathsf{vk}, \mathsf{pid}, \mathsf{S}_i)$.

5. *Resolution Phase:* If the client passed the above checks for S_i then S_i resolves the complaints raised against the client by other servers on $\mathcal{F}_{\mathsf{BC}}$ as follows:

   (a) Parse a complaint by S_k as $(\mathsf{Complaint}, \mathsf{vk}, \mathsf{pid}, \mathsf{S}_k, j, \mathsf{sk}_{\mathsf{ID}})$ where $\mathsf{ID} := (\mathsf{vk}, \mathsf{pid}, \mathsf{S}_k)$.

   (b) Read $(\mathsf{E}_k, \mathsf{csh}_{j,k})$ from $\mathcal{F}_{\mathsf{BC}}$ and decrypt the server's shares as $\mathsf{msg}_k := \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}}, \mathsf{E}_k)$ from E_k and parse $([\![\mathsf{v}_1]\!]_k, [\![\mathsf{r}_1]\!]_k, [\![\mathsf{v}_2]\!]_k, [\![\mathsf{r}_2]\!]_k, [\![\mathsf{v}_3]\!]_k, [\![\mathsf{r}_3]\!]_k, [\![\mathsf{v}_4]\!]_k, [\![\mathsf{r}_4]\!]_k) := \mathsf{msg}_k$.

   (c) Verify the server provided secret key $\mathsf{sk}_{\mathsf{ID}}$ for ID as

   $$\mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}}, \mathsf{Enc}(\mathsf{mpk}, \mathsf{ID}, m)) = m,$$

   where m is randomly sampled from the IBE message space. If the check fails, then discard S_k.

   (d) Discard the client if $\mathsf{csh}_{j,k} \neq \mathsf{Com}([\![\mathsf{v}_j]\!]_k; [\![\mathsf{r}_j]\!]_k)$, else discard S_k.

6. If the client passes the above checks then consider the vote as server-validated.

Once the expiry of a poll is reached, the voting phase has ended. The servers then proceed to perform vote aggregation and output computation.

Vote Aggregation For all the valid clients, each server S_i sums up their own shares as

$$[\![\mathsf{out}_j]\!]_i := \sum_{\text{Valid clients}} [\![\mathsf{v}_j]\!]_i, [\![\mathsf{rout}_j]\!]_i := \sum_{\text{Valid clients}} [\![\mathsf{r}_j]\!]_i \text{ for } j \in [4]$$

Compute the commitments to the output as

$$\mathsf{cout}_j := \prod_{\text{Validated clients}} \mathsf{c}_j \text{ for } j \in [4],$$

where c_j was the client's vote for option j.

Output Computation: The servers reconstruct output as:

1. Each server S_i posts its share of the output - $(\{[\![\mathsf{out}_j]\!]_i, [\![\mathsf{rout}_j]\!]_i\}_{j\in[4]})$ on $\mathcal{F}_{\mathsf{BC}}$.

2. Upon receiving shares from $t+1$ servers in set S perform the following check for each $j \in [4]$:

   $$\mathsf{cout}_j == \mathsf{Com}(\mathsf{out}_j; \mathsf{rout}_j), \text{ where}$$

   $$\mathsf{out}_j := \mathsf{Recon}(0, \{S_k, [\![\mathsf{out}_j]\!]_k\}_{k\in S}), \text{ and}$$

   $$\mathsf{rout}_j := \mathsf{Recon}(0, \{S_k, [\![\mathsf{rout}_j]\!]_k\}_{k\in S}).$$

   If the check passes then proceed to next step. Otherwise, all the servers computes the commitments to the share (of the output) of every other server as:

   $$\mathsf{coutsh}_{j,i} := \prod_{\text{Validated clients}} \mathsf{csh}_{j,i} \text{ for } j \in [4], i \in [n].$$

   Let $B := \emptyset$ denote the set of all the undiscarded servers. Add $B := B \cup \{i\}$ if the share of S_i matches with its commitment for all $j \in [4]$, i.e.

   $$\mathsf{Com}([\![\mathsf{out}_j]\!]_i; [\![\mathsf{rout}_j]\!]_i) == \mathsf{coutsh}_{j,i}.$$

   The servers wait (until $|B| = t+1$) for shares from other servers and repeat the above step. Once $|B| = t+1$ valid shares are identified, reconstruct the outputs $(\mathsf{out}_1, \mathsf{out}_2, \mathsf{out}_3, \mathsf{out}_4)$ as:

   $$\mathsf{out}_j := \mathsf{Recon}(0, \{B_k, [\![\mathsf{out}_j]\!]_k\}_{k\in[B]}) \text{ for } j \in [4],$$

   Compute the opening randomnesses as

   $$\mathsf{rout}_j := \mathsf{Recon}(0, \{B_k, [\![\mathsf{rout}_j]\!]_k\}_{k\in[B]}) \text{ for } j \in [4].$$

3. The servers post $(\mathsf{pid}, \{\mathsf{out}_j, \mathsf{rout}_j, \mathsf{cout}_j\}_{j\in[4]})$ as the output on $\mathcal{F}_{\mathsf{BC}}$.

Private Reconstruction. In the case of private reconstruction towards a specific party P, the servers send their shares to party P. Then party P locally discards the servers whose shares do not match with the commitments. Finally P reconstructs the output by using the undiscarded server shares, and matches it against the publicly computed commitments.

For the sake of brevity, we present the technical details of our NIZK protocols in Section E. Next, we discuss the security properties guaranteed by our above protocol.

1. **Voter Privacy:** The individual votes cast by a client always remain private against a quantum adversary. The vote values are hidden in due to IND-ID-CPA security of the IBE encryption scheme Enc, the perfectly hiding property of Com and perfect zero-knowledge property of the NIZKs.

2. **Censorship Resistance:** A client's vote is guaranteed to be included and verifiable via $\mathcal{F}_{\mathsf{BC}}$. Votes can be censored if a bad server proves a mismatch between an encrypted share and the commitment, which occurs with negligible probability due to IBE's key verifiability.

3. **Public Verifiability:** All commitments and NIZKs are posted on $\mathcal{F}_{\mathsf{BC}}$, allowing any external party to compute output commitments and verify openings and proofs, ensuring public verifiability of the voting process.

4. **Accountability:** The protocol records all actions on $\mathcal{F}_{\mathsf{BC}}$, enabling publicly detecting and identifying any misbehaving parties.

6 Evaluation

We implement the post-quantum and long-lasting PVSS schemes, outlined in sections 4 and 5.1 respectively. Moreover, we use the long-lasting PVSS scheme to build a voting

Servers	Long-lasting PVSS (sec. 5.1)			Post-quantum PVSS (sec. 4)			Gentry et al. [37]		
	Dealing	Verification	Size	Dealing	Verification	Size	Dealing	Verification	Size
16	10.69 ms	5.38 ms	60.5 KB	9.84 ms	4.16 ms	61.3 KB	36.57 s	1.12 s	2868 KB
32	21.96 ms	7.33 ms	121 KB	19.72 ms	4.16 ms	122.5 KB	53.26 s	1.13 s	5804 KB
64	42.51 ms	12.46 ms	242 KB	38.95 ms	4.17 ms	245 KB	100.8 s	1.13 s	11.61 MB
128	85.84 ms	30.48 ms	484 KB	78.71 ms	4.21 ms	490 KB	196.0 s	1.19 s	23.85 MB
256	167.31 ms	114.09 ms	968 KB	159.76 ms	4.29 ms	980 KB	406.6 s	1.20 s	50.0 MB
512	368.46 ms	492.55 ms	1936 KB	296.41 ms	4.28 ms	1960 KB	882.36 s	1.19 s	109.19 MB
1024	867.30 ms	2.13 s	3872 KB	692.35 ms	4.33 ms	3920 KB	2060.48 s	1.25 s	251.5 MB

Table 2: Measurements for Long-lasting and Post-quantum PVSS schemes, in comparison to Gentry et al. [37]

application, as detailed in section 5.2. Our implementation can be found in [53].

Setup: All protocols are implemented in Rust. Elliptic-curve operations (commitments, proofs, and group arithmetic) use `curve25519-dalek` on Ristretto255 [1]. For identity-based encryption (IBE), we use the construction of Ducas et al. [32], which works over NTRU lattices, and its open-source implementation [2][6]. The on-chain vote-verification logic is implemented as an Aptos Move smart contract [3]. All experiments were run on a MacBook Pro with an Apple M4 Max Pro chip and 64 GB RAM using a *single computational thread*. Each reported data point is the mean over 10 independent runs.

6.1 PVSS Experimental Results

Along with the two PVSS schemes from this paper, we also experiment with the lattice-based non-interactive VSS scheme by Gentry et al [37]. We compare all three schemes on the following dimensions: dealer running time, receiver running time, and communication cost. The dealer time includes the time for a dealing client to generate encrypted shares and the proof for all servers. The communication cost is the size of the payload that the dealer *must* publish/broadcast and the pairwise communication. The receiver verification time includes the time to derive the IBE decryption key, decrypt the IBE-encrypted share, and verify the proof.

The empirical measurements in Table 2 clearly demonstrate that interaction between the receivers/verifiers can avoid expensive NIZKs and achieve three orders of magnitude improvement in running time (and memory consumption).

Runtime Comparison. In our post-quantum secure PVSS, the identity-specific secret key derivation accounts for 90-95% of the running time, while the other computations – deserialization, IBE encryptions/decryptions polynomial evaluation – are relatively inexpensive. In contrast, our long-lasting PVSS computes a SCRAPE test during verification, whose running time grows with the server count. The SCRAPE degree-test requires Multi-Scalar Multiplication and that forms a dominant component of the running time. Hence, applications that do not need the homomorphic property can benefit from the post-quantum PVSS, which has a much smaller verification time (apart from being fully post-quantum secure). Meanwhile, the runtime of [37] is dominated by their encryptions, as it requires 1.2 s to compute an encryption. Usage of Bulletproofs, instead of lattice-based NIZK, improved their proving runtime yet incurs a significant overhead (1000×) compared to our post-quantum PVSS. Similar improvements were observed for our PVSS protocols over [37] in terms of memory consumption, which is significant for their scheme.

Communication Comparison. The lattice-based IBE ciphertexts (3840 bytes each) account for most of the communication size in each dealing, in both of our PVSS schemes. The post-quantum PVSS has slightly higher communication cost due to the use of hash-based commitments which grow with the number of servers. In contrast, [37] incurs two orders of magnitude overhead in communication due to the lattice-based PVW encryptions [51] used in their dealing.

6.2 Polling Experimental Results

The various components of the voting application are depicted in Sec. 5.2. The voters use a command-line interface (CLI) app to submit their vote on-chain[7], which includes the IBE-encrypted Shamir shares, commitments, and NIZK proof. Next, the servers decrypt and validate the votes, and perform any complaints as mandated by the PVSS protocol. Finally, the servers aggregate the votes and compute the outcome, as described in section 5.2.

To evaluate scalability for polling application, we systematically varied the number of poll choices $\in \{4, 8, 16\}$, and the number of servers in the committee $n \in \{16, 32, 64, 128\}$. We report costs separately for each role (i) *Client* - Time to produce an encrypted vote and the accompanying proofs, (ii) *Smart Contract* - Time for the Aptos Move contract to ver-

[6]The library [2] is written in C++, so we expose it to Rust via FFI bindings.

[7]Since on-chain storage can be expensive, the system can be easily modified to use some data availability solution.

Poll Choices	Vote Size (KB) # of Servers 16	32	64	128	Smart Contract Cost (ms)
4	258	516	1032	2064	0.452
8	517	1033	2065	4129	0.806
16	1034	2066	4130	8258	1.514

Table 3: Vote sizes for different numbers of poll choices and servers, and the corresponding MOVE smart contract verification time to verify an encrypted vote. The verification time is independent of the number of servers.

ify a submitted vote, and (iii) *Server* - Time for Client vote validation, Votes aggregation, and Output computation.

Client Costs. Figure 1 plots the total time for a poll participant to generate an encrypted vote, including zero-knowledge proofs of correct encryption. Client-side vote encryption and zero-knowledge proof generation increase with the number of choices and servers. For 128 servers, it takes 673 ms and 2.7 s to vote for 4 and 16 poll choices, respectively, showing that client computation dominates latency but remains practical.

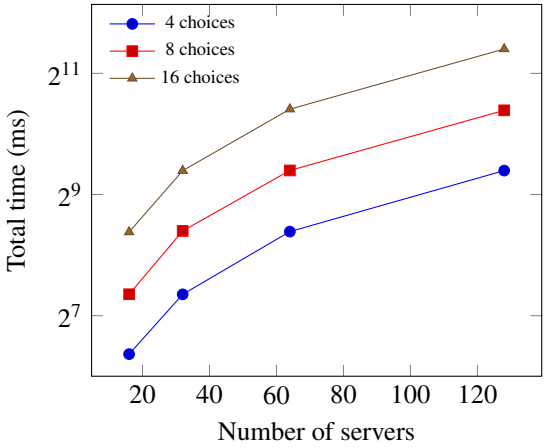
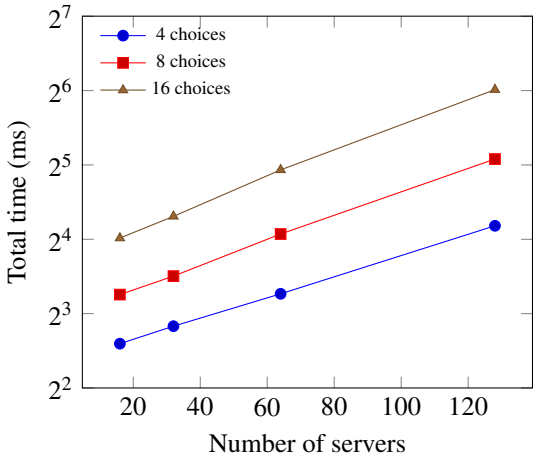


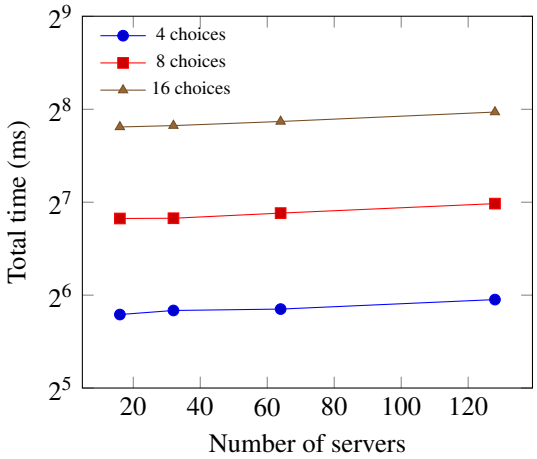
Figure 1: Computation time (ms) required for a poll participant to generate an encrypted vote using our private polling protocol, for varying numbers of poll choices.

Smart Contract Costs. Smart contract verification (reported in Table 3) of encrypted votes scales with the number of choices but not servers, taking under 1.6 ms even for 16 choices, making the on-chain overhead negligible compared to client and server computations.

Server Costs. Each server in the committee validates all client votes and computes the final output. This process involves two phases: (i) *client's vote validation* performed independently for each vote, and (ii) *vote aggregation and output computation* done once collectively over all validated votes. Figure 2a shows that server validation of a single vote scales linearly with poll choices and committee size, remaining under 70 ms



(a) Per-vote validation.



(b) Aggregation and output reconstruction (100,000 votes).

Figure 2: Server computation in our private polling protocol with varying number of poll choices. (a) Per-vote validation performed independently by each server. (b) Committee-wide aggregation and output computation across all validated votes.

even for 128 servers. This is the most expensive protocol step, but it is performed locally by each server whenever a client submits a vote. Figure 2b shows that aggregating 100,000 votes and computing the final output scales mainly with poll choices, increasing only slightly with more servers, and remains under 300 ms even for 128 servers and 16 choices.

Discussion. Our evaluation shows scalable, secure voting with post-quantum vote privacy, low client overhead, a lightweight smart contract, and efficient server-side computation - making it practical for large scale deployment.

7 Conclusion

As PVSS gains attention for its use in decentralized ecosystems like blockchains, it is important to study PVSS compatible with such settings – we began by examining PVSS with non-interactive dealers. Since the lack of post-quantum security hinders adoption of applications like private polling, we focus on post-quantum secure (or at least post-quantum private) PVSS. Our construction improves upon the state-of-the-art post-quantum PVSS results by two orders of magnitude. Along the way, we propose a new model, long-lasting security, which guarantees post-quantum privacy. Exploring schemes in this new model is an exciting direction, especially for blockchain-based applications.

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A Additional Related Works

We compare the everlasting security notion with long-lasting security in this section. Then, we briefly compare our voting scheme with other relevant voting protocols.

A.1 Comparison between Everlasting and Long-lasting Security Models

Everlasting Model: Unruh [61] proposed the notion of everlasting security that requires the protocol to be secure against classical PPT adversaries during protocol execution and secure against computationally unbounded adversaries after protocol execution. They constructed quantum-UC secure communication protocols using trusted signature cards and

quantum channels. Following this work, there has been constructed everlasting secure MPC protocols [9] in the honest majority setting, assuming secure channels. Then [42] constructed certified everlasting secure functional encryption assuming quantum channels. There have been secure voting protocols [41, 52] with everlasting privacy as well. However, all these protocols require quantum channels for secure communication. To implement a secure channel in the everlasting security model, one needs to have quantum secure channels. An alternate solution was proposed by [31] where a secure channel is assumed between one designated party, called the hypervisor, and the protocol participants.

However, the current practical feasibility of quantum channels for regular cryptographic protocols is unclear, and the assumption of a secure channel between the parties and the hypervisor is also not realistic in nature. Hence, everlasting security is difficult to implement in practice. So, we relaxed the notion of everlasting security to long-lasting security where the post-execution security is only required to hold against post-quantum adversaries instead of information-theoretic adversaries. This is a meaningful relaxation as it allows us to build provably secure PVSS (with non-interactive dealer) and voting protocols that can be readily instantiated using post-quantum secure assumptions and highly optimized DL-based cryptography. As an application, we provide a long-lasting secure voting protocol.

A.2 Comparison of Secure Voting Protocols with Post-Quantum Vote Privacy

We compare our voting protocol with other voting protocols that provide post-quantum or everlasting privacy.

Post-Quantum Secure Voting: The state-of-the-art lattice-based voting scheme by [29] enforces the voting client to vote by secret sharing its vote, committing to it and encrypting the shares under each server's public key. The client also proves that the committed shares sum up to 0 or 1. Then each server decrypts its share and computes a valid NIZK proof that the committed share has been tallied correctly. This way, a client delegates the proof computation to the voting servers. Considering 4 servers, each of their server runs for 41 min to conduct a poll with two choices and participation from 11K voters. Meanwhile, our polling smart contract+server protocol performs the same task in a minute, yielding a 40× improvement. We note that this would further increase with the increase in the number of poll choices and the number of servers, as our cost grows sublinearly. The Epoque protocol [19] has similar performance as [29] since they use the same underlying NIZK proofs, and hence our improvement over Epoque would be similar.

Everlasting Secure Voting: There are also several everlasting secure voting protocols [41, 52] based on different properties that a voting scheme can satisfy. We do not delve into that for the sake of brevity. However, we note that our longlasting secure protocol provides similar performance as the state-of-the-art protocol of [52].

B Additional Preliminaries

Here, we present the full details of our preliminaries from Sec. 2 for the sake of completeness.

B.1 SCRAPE-Style Degree Test [23]

Setting. Let $\mathbb{F}$ be a finite field of size q and let $I = \{x_1, \ldots, x_n\} \subset \mathbb{F}$ be distinct, public evaluation points (often $x_i = i$). A dealer claims there is a polynomial $f, g \in \mathbb{F}[X]$ of degree $\deg(f) \le t$ and shares $s_i = f(x_i), \rho_i = g(x_i)$. The dealer publishes *per-share commitments*

$$C_i := \mathsf{Com}(s_i; \rho_i) \in \mathbb{G}, \quad i = 1, \ldots, n,$$

where $\mathsf{Com}(m; r) = g^m h^r$ is a Pedersen commitment in a group $\mathbb{G}$ of order q with generators g, h and unknown $\log_g h$. Pedersen is additively homomorphic: $\prod_i \mathsf{Com}(m_i; r_i)^{\alpha_i} = \mathsf{Com}\big(\sum_i \alpha_i m_i \,;\, \sum_i \alpha_i r_i\big)$.

Definition B.1 (Extended Vandermonde Kernel)**.** Let $V \in \mathbb{F}^{(t+1)\times n+1}$ be the Vandermonde matrix with $V_{j,i} = x_i^j$ for $j = 0, \ldots, t$. Its right kernel is

$$\mathcal{N} = \big\{ r \in \mathbb{F}^{n+1} : \sum_{i=0}^{n} r_i x_i^j = 0 \ \ \forall\, j = 0, \ldots, t \big\},$$

which has dimension $(n+1) - (t+1) = n - t$ when the x_i are distinct. For any $r \in \mathcal{N}$ and any polynomial f with $\deg(f) \le t$, it holds that $\sum_{i=0}^{n} r_i f(x_i) = 0$.

High-level idea. If the posted shares (s_i) do not arise from any deg $\le t$ polynomial, then for a random $r \in \mathcal{N}$ we will have $\sum_i r_i s_i \neq 0$ with high probability. By homomorphism, $D(r) := \prod_i C_i^{r_i} = \mathsf{Com}\big(\sum_i r_i s_i \,;\, \sum_i r_i \rho_i\big)$ is a commitment to $\mathsf{Com}(0; 0)$ iff the shares are t-degree consistent.

The SCRAPE Test $\mathcal{F}_{\mathsf{SCRAPE}}$ over Com. A public basis $N \in \mathbb{F}^{n\times(n-t-1)}$ of $\mathcal{N}$ (depends only on I, t) is fixed once, group $\mathbb{G}$ and the generators g, h. We model a hash $\mathrm{H} : \{0,1\}^* \to \mathbb{F}^{n-t-1}$ as random oracle for Fiat–Shamir.

- *Public Inputs:* I, t; commitments $C_0, C_1, \ldots, C_n \in \mathbb{G}$; basis N of $\mathcal{N}$.
- *Challenge:* $u := \mathrm{H}(I, t, \mathbb{G}, g, h, C_0, C_1, \ldots, C_n) \in \mathbb{F}^{n-t}$; set $r := N \cdot u \in \mathcal{N} \subseteq \mathbb{F}^{n+1}$.
- *Aggregation:* Compute $D := \prod_{i=1}^{n} C_i^{r_i} \in \mathbb{G}$.
- *Local Verification:* Output 1 if $D = 1$, else output 0.

Lemma B.1 (Completeness)**.** *If $s_i = f(x_i)$ and $\rho_i = g(x_i)$ for some polynomials f, g with $\deg(f), \deg(g) \leq t$, then for any $r \in \mathcal{N}$, $\sum_i r_i s_i = 0, \sum_i r_i \rho_i = 0$. Hence, $D = \mathsf{Com}(0;0)$ admits a valid zero-opening with zero-randomness and the verifier accepts.*

Lemma B.2 (Soundness (informal))**.** *Suppose the vectors $s = (s_0, s_1, \ldots, s_n)$ or $\rho = (\rho_0, \rho_1, \ldots, \rho_n)$ are not evaluations of any polynomials of degree $\leq t$. Then for a uniformly random $u \in \mathbb{F}^{n-t}$ (hence $r = Nu \in \mathcal{N}$), $\Pr\left[\sum_i r_i s_i = 0, \sum_i r_i \rho_i = 0\right] \leq 1/q$. Repeating with μ independent challenges drives the soundness error to $\leq 1/q^{\mu}$.*

B.2 Pedersen Commitment [25]:

It has three algorithms:

- $\mathsf{Gen}(1^\lambda)$: Sample $g, h \leftarrow \mathbb{G}$ where $|\mathbb{G}| = O(2^\lambda)$ and set the public parameters as (g, h).
- $\mathsf{Com}(m; s)$: To commit to message m, sample the randomness $s \leftarrow \mathbb{Z}_q$ and compute the commitment as $c = g^m h^s$.
- $\mathsf{Verify}(c, m, s)$: To verify the opening of the commitment check that $c \stackrel{?}{=} g^m h^s$.

It is a perfectly hiding and computationally binding commitment scheme assuming the hardness of solving Discrete-Log.

B.3 Identity-Based Encryption

An Identity-Based Encryption (IBE) scheme [17] is a type of public-key encryption in which the public key of a user can be derived from an arbitrary string representing the user's identity (e.g., an email address).

Formally, an IBE scheme consists of the following probabilistic polynomial-time algorithms:

- $\mathsf{KeyGen}(1^\lambda) \rightarrow (\mathsf{mpk}, \mathsf{msk})$: Takes a security parameter λ and outputs a master public key mpk and a master secret key msk.
- $\mathsf{Extract}(\mathsf{msk}, \mathsf{ID}) \rightarrow \mathsf{sk}_{\mathsf{ID}}$: Takes the master secret key msk and an identity string ID, and outputs the private key $\mathsf{sk}_{\mathsf{ID}}$ corresponding to ID.
- $\mathsf{Enc}(\mathsf{mpk}, \mathsf{ID}, m) \rightarrow c$: Takes the master public key mpk, an identity ID, and a message m, and outputs a ciphertext c encrypted under ID.
- $\mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}}, c) \rightarrow m$: Takes the private key $\mathsf{sk}_{\mathsf{ID}}$ and a ciphertext c, and outputs the original message m.
- $\mathsf{VerKey}(\mathsf{mpk}, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}}) \rightarrow \{0, 1\}$: Takes the master public key mpk, identity ID and a identity secret key $\mathsf{sk}_{\mathsf{ID}}$ and outputs 1 if $\mathsf{sk}_{\mathsf{ID}}$ is the identity secret key corresponding to ID otherwise outputs 0.

Correctness: An IBE scheme is said to be **correct** if, for every identity string ID and every message m in the message space, the decryption algorithm correctly recovers m from the ciphertext c produced by the encryption algorithm. Formally, for all ID and m:

$$\Pr[\mathsf{Dec}(\mathsf{Extract}(\mathsf{msk}, \mathsf{ID}), \mathsf{Enc}(\mathsf{mpk}, \mathsf{ID}, m)) \neq m] \leq \mathsf{negl}(\lambda).$$

IND-ID-CPA Security: An IBE scheme is said to be indistinguishable under chosen plaintext attack for a given identity (IND-ID-CPA) if a PPT adversary $\mathcal{A}$ wins the following game with $\mathsf{negl}(\lambda)$ probability.

<u>IND-ID-CPA Game</u>:

1. The challenger runs $\mathsf{KeyGen}(1^\lambda) \rightarrow (\mathsf{mpk}, \mathsf{msk})$ and gives mpk to $\mathcal{A}$.
2. $\mathcal{A}$ may query the key extraction oracle $\mathsf{Extract}(\mathsf{msk}, \mathsf{ID})$ for any identity ID to receive $\mathsf{sk}_{\mathsf{ID}}$.
3. The adversary chooses two messages m_0, m_1 and a challenge identity ID^* that has not been queried to the $\mathsf{Extract}(\mathsf{msk}, \cdot)$ oracle.
4. The challenger flips a bit $b \leftarrow \{0, 1\}$ and returns $c^* \leftarrow \mathsf{Enc}(\mathsf{mpk}, \mathsf{ID}^*, m_b)$.
5. The adversary continues to make key extraction queries for $\mathsf{ID} \neq \mathsf{ID}^*$ and eventually outputs a guess b'.

The IBE scheme is IND-ID-CPA secure if:

$$\left|\Pr[b' = b] - \frac{1}{2}\right| \leq \mathrm{negl}(\lambda)$$

We use the lattice-based IBE scheme of [32] for our protocol. It is based on the GPV scheme [38] over NTRU lattices and it is post-quantum secure under the Ring-LWE [48] assumption.

Key Verifiability: An IBE scheme is said to satisfy key verifiability if for any security parameter λ, any identity ID, any message m, and any stateful PPT adversary $\mathcal{A}$, the following probability is negligible in λ.

$$\Pr\left[\begin{array}{c|c} \mathsf{VerKey}(\mathsf{mpk}, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID}}) = 1; & \mathsf{mpk} \leftarrow \mathcal{A}(1^\lambda); \\ m \neq \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID}}, c) & c \leftarrow \mathsf{Enc}(\mathsf{mpk}, \mathsf{ID}, m); \\ & \mathsf{sk}_{\mathsf{ID}} \leftarrow \mathcal{A}(c) \end{array}\right]$$

Informally, except with negligible probability, an adversary can not produce a master public key, such that it can later release a decryption key which verifies correctly, but leads to an incorrect decryption.

The Boneh-Franklin IBE scheme [17] and the lattice-based IBE scheme [32] satisfy this property. In the security proofs of our PVSS protocols, we also argue the adaptive security of the schemes. For that, we need the IBE encryptions to be non-committing [22], i.e., a simulated ciphertext can be later opened consistently as an encryption of any message.

Non-Committing Encryption: The IBE in [32] is non-committing in the programmable random oracle model since it encrypts a random key k using the IBE and then encrypts the message m as $\mathrm{H}(k)+m$, where H is the random oracle. The ciphertext can be programmed to open to any message m by programming $\mathrm{H}(k)$. An adversary is able to prevent this programming if it decrypts k from the IBE scheme and queries it to the H function. However, this breaks the security of the encryption scheme. Hence, adaptive security of their IBE reduces to the IND-ID-CPA security of the IBE in the random oracle model.

Post Quantum Secure IBE We say that an IBE scheme is post-quantum secure if all of the above properties hold against an adversary with quantum capabilities. The lattice-based scheme by Gentry et al. [32] is post-quantum secure under the Ring-LWE [48] assumption.

B.4 Shamir's Secret Sharing Scheme

We recall the definition of an (n,t) Shamir secret sharing scheme as follows:

Definition B.2 (Shamir's Secret Sharing Scheme)**.** Let $\mathbb{F}$ be a finite field. It consists of two algorithms:

- $\mathsf{Share}(s,n,t)$: The dealer selects a random t-degree polynomial

$$f(x) := s + a_1x + a_2x^2 + \cdots + a_tx^t \in \mathbb{F}[x],$$

  where the coefficients $a_1,\ldots,a_t \in \mathbb{F}$ are chosen uniformly at random. For each party $i \in [n]$, the dealer computes the share $[\![s]\!]_i = f(i) \in \mathbb{F}$. The dealer outputs $(\{[\![s]\!]_i\}_{i\in[n]})$.

- $\mathsf{Recon}(z,\{i_j,[\![s]\!]_{i_j}\}_{j=1}^{t+1})$: Given any $t+1$ shares, Recon reconstructs $f(z)$ as:

$$f(z) = \sum_{j=1}^{t+1} [\![s]\!]_{i_j} \cdot \lambda_j,$$

  where $\lambda_j(z) = \prod_{\substack{1\le m\le t+1\\ m\neq j}} \frac{z-i_m}{i_j-i_m}$ are the Lagrange interpolation coefficients evaluated at $x=z$.

- $\mathsf{ReconPoly}(\{i_j,[\![s]\!]_{i_j}\}_{j=1}^{t+1})$: Given any $t+1$ shares, ReconPoly reconstructs the polynomial $f(x)$ as:

$$f(x) = \sum_{j=1}^{t+1} [\![s]\!]_{i_j} \cdot \lambda_j,$$

  where $\lambda_j(x) = \prod_{\substack{1\le m\le t+1\\ m\neq j}} \frac{x-i_m}{i_j-i_m}$.

The scheme satisfies the following properties:

- **Correctness:** Any set of $t+1$ or more valid shares correctly reconstructs the secret s.
- **Perfect Secrecy:** Any set $\le t$ shares information theoretically hides s.

Shamir sharing scheme also satisfies reconstruction over additively homomorphic commitment schemes defined over message space field $\mathcal{M}$ and randomness space field $\mathcal{R}$ as follows:

- $\mathsf{ShareCom}(n,t)$: Samples a **random secret** $s \in \mathcal{M}$ and randomness $r \in \mathcal{R}$, the dealer selects two random t-degree polynomial

$$f(x) := s + a_1x + a_2x^2 + \cdots + a_tx^t \in \mathcal{M}[x],$$

$$g(x) := r + b_1x + b_2x^2 + \cdots + b_tx^t \in \mathcal{R}[x]$$

  where $a_1,\ldots,a_t \in \mathcal{M}$ and $b_1,\ldots,b_t \in \mathcal{R}$ are chosen uniformly at random. For each party $i \in [n]$, the dealer computes the share $[\![s]\!]_i := f(i) \in \mathcal{M}$ and $[\![r]\!]_i := g(i) \in \mathcal{R}$, and the committed share $\langle [\![s]\!]_i; [\![r]\!]_i\rangle$. The dealer outputs $(s,r,\{[\![s]\!]_i,[\![r]\!]_i,\langle [\![s]\!]_i;[\![r]\!]_i\rangle\}_{i\in[n]})$.

- $\mathsf{ShareCom}(s,n,t)$: Given a **secret** $s \in \mathcal{M}$ and randomness $r \in \mathcal{R}$, the dealer selects two random t-degree polynomial

$$f(x) := s + a_1x + a_2x^2 + \cdots + a_tx^t \in \mathcal{M}[x],$$

$$g(x) := r + b_1x + b_2x^2 + \cdots + b_tx^t \in \mathcal{R}[x]$$

  where $a_1,\ldots,a_t \in \mathcal{M}$ and $b_1,\ldots,b_t \in \mathcal{R}$ are chosen uniformly at random. For each party $i \in [n]$, the dealer computes the share $[\![s]\!]_i := f(i) \in \mathcal{M}$ and $[\![r]\!]_i := g(i) \in \mathcal{R}$, and the committed share $\langle [\![s]\!]_i; [\![r]\!]_i\rangle$. The dealer outputs $(s,r,\{[\![s]\!]_i,[\![r]\!]_i,\langle [\![s]\!]_i;[\![r]\!]_i\rangle\}_{i\in[n]})$.

- $\mathsf{ReconCom}(z,\{i_j,\langle [\![s]\!]_{i_j};[\![r]\!]_{i_j}\rangle\}_{j=1}^{t+1})$: Given any $t+1$ shares, Recon reconstructs the committed secret $\langle s;r\rangle$:

$$\langle s;r\rangle = \prod_{j=1}^{t+1} \langle [\![s]\!]_{i_j};[\![r]\!]_{i_j}\rangle^{\lambda_j(z)}.$$

C Long-lasting Security Model

A protocol is said to be long-lasting secure if it is secure against a PPT classical adversary *during* the protocol execution. And after the protocol terminates, the protocol transcript would remain secure (does not leak any information about honest parties' inputs) against a future PPT quantum adversary.

Real-World: In the real-world execution, the protocol π is carried out between the honest parties $\{P_i\}_{i\in[n]}$ and the real-world adversary $\mathcal{A}$. All the parties are classical PPT Turing machines, and $\mathcal{A}$ has auxiliary information z. At the outset of the protocol, each party P_i has its own input x_i. At the outset of the protocol, $\mathcal{A}$ may or may not corrupt a party. Upon corruption of a party, $\mathcal{A}$ gets access to the internal state

and input of that party. From now on, the party will behave according to $\mathcal{A}$'s instructions (since we are in the malicious model). At the end of the protocol, the honest parties produce their outputs. The view of $\mathcal{A}$ consists of the internal states of the corrupt parties and their outputs. We denote the real-world adversarial view as $\mathsf{Real}_{\pi,\mathcal{A},}(1^\lambda, z)$, and it consists of the honest parties' outputs, the corrupt parties' outputs, and the internal states of the corrupt parties.

Ideal-World: In the ideal-world execution, we consider the honest parties $\{P_i\}_{i\in[n]}$, the ideal-world adversary Sim and the functionality $\mathcal{F}$. Sim has access to the real-world adversarial algorithm $\mathcal{A}$ and the protocol π. All the parties are classical PPT Turing machines. Sim has a random tape r and security parameter λ. It simulates the role of $\mathcal{A}$ in the ideal world, and whenever $\mathcal{A}$ corrupts a party in the real world Sim corrupts that party in the ideal world and gets access to its internal state. Sim invokes the algorithm of $\mathcal{A}$, in his head, in another internal protocol execution where Sim simulates the view of the honest parties to $\mathcal{A}$. We will denote this internal copy of $\mathcal{A}$ as $\mathcal{A}_{\mathsf{Int}}$. Based on the reply of $\mathcal{A}_{\mathsf{Int}}$ in the internal execution, Sim behaves accordingly in the ideal world execution. He extracts the inputs of the corrupted parties in the internal execution and invokes $\mathcal{F}$ in the ideal world with those inputs to obtain the output. In the internal execution he simulates the protocol in such a way that $\mathcal{A}_{\mathsf{Int}}$ obtains that output. At the end of the protocol, the view of $\mathcal{A}_{\mathsf{Int}}$ consists of the internal states of the corrupt parties and their outputs. We denote the ideal-world adversarial view as $\mathsf{Ideal}_{\mathcal{F},\mathsf{Sim}}(1^\lambda, z)$, and it consists of the honest parties' outputs, the corrupt parties' outputs, and the internal states of the corrupt parties.

Definition C.1. Let π be a protocol for computing a functionality $\mathcal{F}$. We say that π computes functionality $\mathcal{F}$ with long-lasting security in the presence of static adversaries if for every PPT classical static real-world adversary $\mathcal{A}$, there exists a PPT classical ideal-world adversary Sim, such that:

$$\mathsf{Real}_{\pi,\mathcal{A}}(1^\lambda, z) \overset{\mathsf{pq}}{\approx} \mathsf{Ideal}_{\mathcal{F},\mathsf{Sim}}(1^\lambda, z).$$

D Security Proofs

In this section, we provide the security proofs of our PVSS protocols.

D.1 Security Proof of Theorem 4.1

We present the security proof of our post-quantum secure PVSS protocol as follows.

Proof. **Correctness** of the decrypted shares follows from the correctness of the Shamir secret sharing scheme and the correct decryption of the IBE scheme. If the dealer is honest, then each $[\![s]\!]_i$ and $[\![r]\!]_i$ is a correct sharing of s and r and the evaluations of $B(i)$ are committed for $i \in [n]$ where $W(x) := B(x) - \mathsf{ch} \cdot f(x)$ and $[\![s]\!]_i := f(i)$. Each receiver checks that the hash of $W(i) + \mathsf{ch} \cdot [\![s]\!]_i$ is the same as the hash of $B(i)$ under randomness $q(i)$, and this always holds.

The PVSS satisfies **non-interactive dealing** since the dealer does not need to participate in the PVSS once it has dealt the shares non-interactively using the Share algorithm.

The PVSS transcript is also **publicly verifiable** in the random oracle model, as any external party can publicly verify a raised complaint $(\textsc{Complaint}, i, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID},i})$ by decrypting $([\![s]\!]_i, [\![r]\!]_i, q_i) := \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID},i}, \mathsf{E}_i)$ and checking it against the proof π.

Reconstructability follows, because if the verification succeeds, then the only way an invalid dealing goes undetected is if the dealer encrypts shares which does not correspond to a valid t-degree sharing of a polynomial, and yet none of the $n-t$ honest receivers raised a complaint. This only occurs if $\mathrm{H}(W(i) + \mathsf{ch} \cdot \widetilde{[\![s]\!]_i}, \widetilde{q}_i) == \mathrm{H}(B(i), q_i)$ where $(\widetilde{[\![s]\!]_i}, \widetilde{q}_i)$ are sent by the malicious dealer to the i-th honest receiver and $\widetilde{[\![s]\!]_i} \neq [\![s]\!]_i$. This requires the adversarial dealer to either predict ch or find a collision in H, both of which occur with negligible probability when H is modeled as a random oracle. All the n shares are committed, and whenever $t+1$ receivers provide their openings $\{i_j, [\![s]\!]_{i_j}, [\![r]\!]_{i_j}, q_{i_j}\}_{j\in[t+1]}$ it can be checked that the provided openings are correct against the commitments.

For **liveness**, the adversary is allowed to corrupt upto $t < \frac{n}{2}$ since reconstructing the secret requires $t+1$ shares from honest receivers.

It can be observed that the dealing remains hidden from a set of t adversarial parties as the dealer uses a t-degree sharing and the adversary lacks one share. The commitments csh_i, c and cpt_i hide the underlying committed shares since H is a random oracle, and so it is preimage resistance. And the encryptions E_i hide the values $([\![s]\!]_i, [\![r]\!]_i, q_i)$ due to the IND-ID-CPA Security of the IBE scheme. The commitments are instantiated using the quantum random oracle and the IBE is post-quantum secure, and hence the scheme achieves **post-quantum secrecy**. □

On Adaptive Security: Our PVSS protocol can be proven to be adaptively secure by assuming the random oracle is programmable and the IBE is non-committing by using a programmable random oracle. In our PVSS, the simulator simulates an honest dealing by sampling two random t-degree polynomials for sharing a dummy secret and randomness r, and committing the shares as csh_i using the random oracle and encrypting those shares using the IBE. When the i-th receiver gets corrupted, the simulator obtains its share and the simulator equivocates csh_i to open the share by programming

the oracle and programming the IBE to open to it. Equivocation succeeds since $[\![r]\!]_i$ remains hidden from the adversary without corrupting the i-th party and hence the simulator can successfully equivocate csh_i. The same argument holds for equivocating the commitment cpt_i and its opening $q(i)$. The encryptions are equivocated based on the non-committing property of the IBE.

D.2 Security Proof of Theorem 5.1

We present the security proof of our long-lasting secure PVSS protocol as follows.

Proof. **Correctness** of the decrypted shares follows from the correctness of the Shamir secret sharing scheme and the correct decryption of the IBE scheme. If the dealer is honest, then each $[\![s]\!]_i$ and $[\![r]\!]_i$ is a correct sharing of s and r, and each i-th commitment csh_i is a commitment to $[\![s]\!]_i$ using randomness $[\![r]\!]_i$. Each receiver checks that the commitments pass the SCRAPE test, using the homomorphic property of Com and the correctness of the SCRAPE test, and the commitments correctly decommit to $[\![s]\!]_i$ and $[\![r]\!]_i$ by relying on the correctness of the commitment.

The PVSS satisfies **non-interactive dealing** since the dealer does not need to participate in the PVSS once it has dealt the shares non-interactively using the Share algorithm.

The PVSS transcript is also **publicly verifiable** in the random oracle model, as any external party can publicly verify a raised complaint $(\textsc{Complaint}, i, \mathsf{ID}, \mathsf{sk}_{\mathsf{ID},i})$ by decrypting $([\![s]\!]_i, [\![r]\!]_i) := \mathsf{Dec}(\mathsf{sk}_{\mathsf{ID},i}, \mathsf{E}_i)$ and checking it against the provided commitment csh_i to server's share.

Recontructability follows, because if the dealing passes the SCRAPE test, then the only way an invalid dealing goes undetected is if the commitments are not to a degree t-sharing, or the dealer encrypts shares that do not correspond to a valid t-degree sharing of a polynomial, and yet none of the $n-t$ honest receivers raised a complaint. The former case occurs with negligible probability due to the soundness of the SCRAPE test B.1. The second case occurs only if the commitments pass the SCRAPE test, but the dealer opens the commitments to shares which do not fall on the same degree t polynomial. However, this requires breaking the binding of the commitment scheme, which occurs with negligible probability.

For **liveness**, the adversary is allowed to corrupt upto $t < \frac{n}{2}$ since reconstructing the secret requires $t+1$ shares from honest receivers.

It can be observed that the dealing remains hidden from a set of t adversarial parties as the dealer uses a t-degree sharing and the adversary lacks one share. The commitments csh_i and c perfectly hide the underlying committed shares. And the encryptions E_i hide the values $([\![s]\!]_i, [\![r]\!]_i)$ due to the IND-ID-CPA Security of the IBE scheme. Hence, the scheme achieves **longlasting secrecy** as the encryptions remain post-quantum secure in the future and commitments are perfectly hiding. □

On Adaptive Security: Our PVSS protocol can be proven to be adaptively secure by assuming that Com is adaptively secure and the IBE is non-committing (by using a programmable random oracle). In our PVSS, the simulator simulates an honest dealing by sampling two random t-degree polynomials for sharing a dummy secret and randomness r, and committing the shares as csh_i using Com and encrypting those shares using the IBE. When the i-th receiver gets corrupted, the simulator obtains its share and the simulator equivocates csh_i to open the share by equivocating Com and equivocating the IBE to open to it and the equivocated randomness. Equivocation succeeds due to the adaptive security of Com and the non-committing property of the IBE.

E Additional Cryptographic Details

This section presents the detailed NIZK protocols used in the polls. We use Pedersen Commitment to instantiate Com. We recall them below.

E.1 Proof of Correct Commitment to the Vote

The client proves that it voted 1 for a single option (among four options) and zero for the other options by proving the following:

- **Input:** $(\mathsf{c}_1, \mathsf{c}_2, \mathsf{c}_3, \mathsf{c}_4, \mathsf{vk}, \mathsf{pid})$
- **Witness:** $(\mathsf{v}_1, \mathsf{r}_1, \mathsf{v}_2, \mathsf{r}_2, \mathsf{v}_3, \mathsf{r}_3, \mathsf{v}_4, \mathsf{r}_4)$
- **Statement:** $\forall j \in [4], (\mathsf{c}_j = \mathsf{Com}(\mathsf{v}_j; \mathsf{r}_j))$ & $(\mathsf{c}_j = \mathsf{Com}(0; \mathsf{r}_j) \vee \mathsf{c}_j = \mathsf{Com}(1; \mathsf{r}_j))$ & $\mathsf{c}_1 + \mathsf{c}_2 + \mathsf{c}_3 + \mathsf{c}_4 = \mathsf{Com}(1)$
- $\mathsf{Prove}_{\mathsf{vote}}$: The prover runs the following:
  1. For each $j \in [4]$: Run $\pi_j := \mathsf{Prove}_{\mathsf{set}}((\mathsf{c}_j, \{0,1\}), (\mathsf{v}_j, \mathsf{r}_j))$.
  2. Sample $c' := \mathsf{Com}(0; r')$. Run $\pi' := \mathsf{Prove}_{\mathsf{set}}((C', \{0\}), (0, r'))$.
  3. Compute $R := \sum_{j \in [4]} \mathsf{r}_j + r'$.
  4. Compute proof $\pi_{\mathsf{vote}} := (\{\pi_j\}_{j \in [4]}, c', R)$.
- $\mathsf{Verify}_{\mathsf{vote}}$: The verifier runs the following:
  1. For each $j \in [4]$: Check $\mathsf{Verify}_{\mathsf{set}}((\mathsf{c}_j, \{0,1\}), \pi_j) = 1$.
  2. Check $\mathsf{Verify}_{\mathsf{set}}((c', \{0\}), \pi') = 1$.
  3. Check $\mathsf{Com}(1; R) = \Pi_{j \in [4]} \mathsf{c}_j \cdot c'$.

If any of the above checks fail then output 0, otherwise, output 1.

Verifying the above three conditions suffices to check that the vote is correctly committed. The first condition ensures that each commitment is a bit commitment. The second and third checks ensure that the sum of all the committed bits is 1. Combining the two arguments, we ensure that only one of the four options was voted with a 1, and the other three were voted 0. We use the proof system from Appendix. E.2 to instantiate the set membership proof.

E.2 Proof of Set membership of a Committed Element

We discuss the OR proof [18] on top of Pedersen commitments to perform this set membership proof in a set X. It is run by a client, identified by verification key vk, in poll identified by pid as follows. The same NIZK proof works for proving that a committed value using a Pedersen commitment is a bit, and also to prove that a commitment is to 1. These can be accommodated by changing the entries of the set X. The proof details are as follows:

- **Input:** Commitment to x as C and a set $X = \{x_1, \ldots, x_N\}$.
- **Witness:** x_j and commitment randomness r.
- **Statement:** Output 1 if $x_j \in X$ and $C = \langle x_j; r \rangle = g^{x_j} h^r$.
- $\mathsf{Prove}_{\mathsf{set}}$: To compute the proof perform the following:
  - Sample simulated responses $z_1, \ldots, z_{j-1}, z_{j+1}, \ldots, z_N \leftarrow \mathbb{Z}_q$ and $w_1, \ldots, w_{j-1}, w_{j+1}, \ldots, w_N \leftarrow \mathbb{Z}_q$. Sample simulated challenges $\mathsf{ch}_1, \ldots, \mathsf{ch}_{j-1}, \mathsf{ch}_{j+1}, \ldots, \mathsf{ch}_{N-1} \leftarrow \mathbb{Z}_q$.
  - For $i \in [N] \setminus j$, compute simulated first messages as $T_i = \frac{g^{z_i} h^{w_i}}{C^{\mathsf{ch}_i}}$.
  - Sample first message randomness for true statement j as $u_j \leftarrow \mathbb{Z}_q, v_j \leftarrow \mathbb{Z}_q$. Compute first message for statement j as $T_j = g^{u_n} h^{v_n}$.
  - Compute the proof challenge $\mathsf{ch} = H(\mathsf{vk}, \mathsf{pid}, g, h, C, X, T_1, \ldots, T_N)$.
  - Compute $\mathsf{ch}_j = \mathsf{ch} - \sum_{i \in [N] \setminus j} \mathsf{ch}_i$.
  - Compute opening for jth statement as $z_j = u_j + \mathsf{ch}_j \cdot x_j$ and $w_j = v_j + \mathsf{ch}_j \cdot r$.

  Output proof as $(\{T_i, \mathsf{ch}_i, z_i, u_i\}_{i \in [N]})$.
- $\mathsf{Verify}_{\mathsf{set}}$: To verify the proof, perform the following:
  - Compute the proof challenge $\mathsf{ch} = H(\mathsf{vk}, \mathsf{pid}, g, h, C, X, T_1, \ldots, T_N)$.
  - Verify $\mathsf{ch} \stackrel{?}{=} \sum_{i \in [N]} \mathsf{ch}_i$.
  - For $i \in [N]$, verify that $g^{z_i} h^{w_i} \stackrel{?}{=} T_i \cdot C^{\mathsf{ch}_i}$.

  If any of the above checks fail, then output 0, otherwise, output 1.

The above scheme is sound assuming Discrete Log is hard, and it satisfies perfect zero-knowledge assuming H is a programmable random oracle model. We refer to Chapter 19 of [18] for the proof details.